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Convergence of Asynchronous Stochastic Gradient Descent for Polyak–Łojasiewicz Functions

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Abstract. We analyze the convergence properties of Asynchronous Stochastic Gradient Descent (A-SGD) by explicitly incorporating the delay into the corresponding stochastic differential equation. Unlike prior approaches that rely on delay-compensation techniques or approximate stochastic differential delay equations, we propose a framework where delays are directly embedded into the optimization process. We establish exponential convergence guarantees under the Polyak–Lojasiewicz (PL) condition and derive a novel bound on steady-state behavior, demonstrating that A-SGD remains stable under realistic delay conditions. We validate our framework through experiments on synthetic functions satisfying the PL condition, obtaining tighter upper bounds than existing methods. We further confirm the theoretical upper bound on an example of an over-parameterized neural network.

Key words: Asynchronous Stochastic Gradient Descent, Polyak-Lojasiewicz functions, Delay compensation

1. Introduction

Many modern machine learning tasks involve extremely large datasets and models, which are too big or too time-consuming for a single computer to handle efficiently. A common solution is to split the work across multiple computational units, often called *workers*. Each worker receives some subset of the data or a portion of the computational task. By collaborating, the workers aim to find a parameter set that minimizes an objective function f. Formally, consider the problem of minimizing a function f, where $\theta \in \mathbb{R}^k$ are the parameters:

$$f(\theta) = \mathbb{E}_{x\sim\mathcal{D}}[f(\theta; x)].$$

Since the expectation is generally intractable (e.g., due to a large or infinite dataset $\mathcal{D}$), we approximate it using a mini-batch of samples $\{x_i\}_{i=1}^{B}$. The standard Stochastic Gradient Descent (SGD)

update is:

$$\theta_{t+1} = \theta_t - \eta_t \cdot \nabla_\theta \left(\frac{1}{B} \sum_{i=1}^{B} f(\theta_t; x_i) \right).$$

In synchronous SGD, we parallelize the computation using N workers. Each worker performs the following steps at iteration t: it receives the current parameters θ_t, which we call $\theta_t^{(j)}$. Next, worker j samples a mini-batch $\mathcal{B}_j$ and computes the *local gradient*:

$$g_t^{(j)} = \nabla_\theta \left(\frac{1}{|\mathcal{B}_j|} \sum_{x \in \mathcal{B}_j} f(\theta_t^{(j)}; x) \right).$$

Once all workers have computed their local gradients, they synchronize and compute the average gradient:

$$\bar{g}_t = \frac{1}{N} \sum_{j=1}^{N} g_t^{(j)}$$

The final step involves updating the model parameters θ:

$$\theta_{t+1} = \theta_t - \eta_t \cdot \bar{g}_t$$

Note that in a *synchronous* training setup, each worker j computes the gradient of f with respect to its parameter vector $\theta_t^{(j)} \in \mathbb{R}^k$, evaluates it on its local data, and then waits until *all* other workers have finished before aggregating their updates to form the new global parameter θ_{t+1}. Only after this joint update is completed does the next round of gradient computation begin. This synchronization mechanism guarantees that the local parameter $\theta_t^{(j)}$ of each worker coincides with the global parameter θ_t at time t. However, such rigid synchronization can be inefficient: if a single worker is delayed–due to slower hardware or network congestion–all other workers must remain idle until it completes.

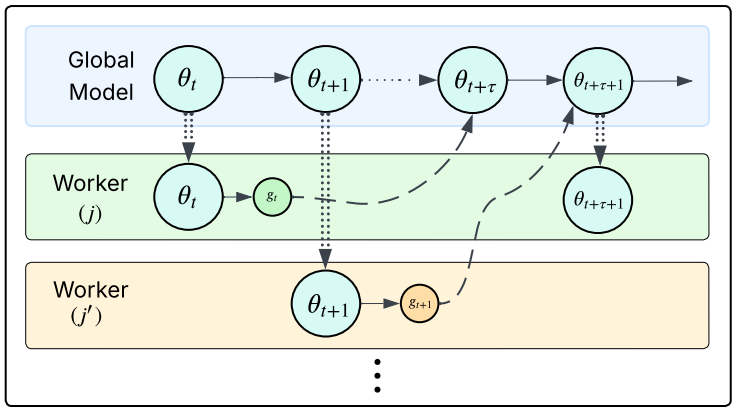


FIGURE 1. A-SGD training process as discussed in [30]†.

To mitigate this bottleneck, one can use A-SGD. In A-SGD, each worker computes its local gradient as soon as it finishes its current batch and sends the update to a central server (or parameter store) without waiting for others. This lets faster workers continue contributing updates immediately. However, it also means some workers may compute gradients using stale versions of the model. For example, a worker might base its gradient on the parameter θ_t at time t, but by the time the update is applied, the global model may already have advanced to $\theta_{t+\tau}$ for some delay τ.

Formally, let g_t be the gradient a worker computes using the model snapshot at time t (see Figure 1). Because of asynchrony, the global model may have moved to $\theta_{t+\tau}$ before g_t is finally applied. This *delayed gradient* phenomenon makes the analysis of A-SGD more challenging than the synchronous case, because updates are no longer calculated from the *current* global model. The main goal of this work is to incorporate these inevitable delays into a modified framework–using an SDE–so that we can rigorously analyze and compensate for their impact: *Under reasonable delay assumptions, and with a suitable learning rate, we prove convergence of A-SGD in expectation to a neighborhood of the optimum, up to a persistent constant that scales at most quadratically with the delay.*

2. Previous Work The continuous-time perspective of SGD [26] has inspired subsequent studies of delay effects and compensation in asynchronous optimization. For instance, [30] proposed an A-SGD algorithm with delay compensation, showing that careful adjustments can mitigate adverse effects of stale gradients. In a related direction, [19] examined noise-to-state stability in stochastic differential equations with persistent noise. The lock-free SGD framework *Hogwild!*, introduced in [23], pioneered subsequent research on asynchronous methods. An in-depth survey on delay and asynchrony in large-scale machine learning can be found in [4]. Additionally, real-world issues related to delayed updates in distributed deep networks are outlined in [6].

Our work is related to the convergence properties of A-SGD under delays. In [8], the error-runtime trade-off is examined, and the *K-async* algorithm and its variant *K-batch-async* are proposed, achieving balanced staleness and iteration speed. By relaxing the standard bounded-delay assumptions, recent work has identified conditions under which asynchronous methods can surpass synchronous counterparts in wall-clock efficiency. In a similar direction, [20] revisits the convergence of A-SGD with staleness, proving that for both convex and nonconvex objectives, convergence rates can be

[†] Before a worker wants to add the gradient g_t (based on the model at time t) to the global model, several other workers may already have added their gradients and the global model has been updated to $\theta_{t+\tau}$, where τ is the delay.

made independent of the maximum delay, depending only on the number of parallel workers. The analysis introduces a virtual iterate framework and employs delay-adaptive step sizes, showing that A-SGD can often outperform equivalent synchronous minibatch SGD in wall-clock time. Finally, in [18] *Ringmaster ASGD* is proposed, which is a fully asynchronous scheme that filters out overly stale gradients via a delay threshold R and applies updates immediately on "fresh-enough" gradients, effectively prioritizing fast workers while disregarding outdated information. The authors prove it is the first A-SGD with optimal time complexity under arbitrary heterogeneous worker speeds–matching known lower bounds–thereby giving wall-clock optimality guarantees rather than iteration-only rates. This establishes a modern theoretical foundation that complements results under the Polyak–Lojasiewicz condition by removing overly pessimistic delay constraints.

In discrete time, [13] studied distributed and federated learning to show improved convergence of $O(\sigma^2\epsilon^{-2} + \sqrt{\tau_{\max}\tau_{\text{avg}}}\epsilon^{-1})$, where σ^2 is the noise variance, $\tau_{\max}$, τ_{avg} are the maximum and average delay, respectively, and ϵ is the suboptimality gap. Refining the bounds in [5], [9] establishes an improved rate of $O(L\tau_{\text{avg}}/\epsilon + \sigma^2\epsilon^{-2})$ for L-smooth convex functions and further derives sharper iteration complexity bounds for proximal incremental aggregated gradient methods. Meanwhile, [15] established a rate of $O(1/\sqrt{K})$ under bounded delay (where K is the number of iterations), and [29] extended this result to unbounded delays.

Another direction is the *fixed-time gradient dynamics* framework for continuous-time optimization [21], where time-varying coefficients enable convergence within a predetermined horizon, independent of initial conditions. These ideas motivate the incorporation of delay-awareness into stochastic optimization. Likewise, [28] explored block-coordinate descent under the Polyak–Lojasiewicz condition in discrete time.

A separate line of work [10] used stochastic differential delay equations (SDDEs) to approximate A-SGD, mainly bounding discretization errors and deriving rates via moment and energy analyses. Our approach departs from such approximations by embedding delay directly into the SDE, removing the need for auxiliary compensation techniques. Under the Polyak–Lojasiewicz condition, we prove exponential convergence and clarify the role of noise and learning rate. Unlike [10], which links noise to batch sizes and delays, our framework highlights their direct proportionality to the learning rate, yielding clearer insight into optimization stability.

On a completely different note, [7] addresses generalization in asynchronous training. Specifically, the authors prove non-vacuous generalization bounds for A-SGD under much weaker conditions than Lipschitz continuity. They establish on-average model stability results for A-SGD and

quantify how factors such as delay, initialization, and sample size affect the excess generalization error. In [16], the authors explore fully asynchronous Local-SGD in large-scale language model training. The authors show that naive asynchronous training can suffer slower convergence due to stale gradients with momentum. They propose a delayed Nesterov momentum correction and adaptive local step schedules to counteract staleness.

3. Our Contributions In this work, a continuous-time stochastic framework is developed to model A-SGD with delayed updates. By embedding gradient delays directly into the SDE–instead of relying on classical SDDEs or discrete compensation techniques–the approach naturally captures the role of delays and the accumulation of noise. Under the Polyak–Lojasiewicz (PL) condition, this formulation yields an exponential convergence bound that quantifies how delays, learning rates, and stochastic noise jointly affect convergence to a steady-state neighborhood of the optimum. The analysis further reveals a direct linear dependence between the steady-state noise level and the learning rate, underscoring the practical trade-off between faster early-stage convergence and a larger long-term error floor. In contrast to prior treatments that impose rigid assumptions on bounded delays or gradient variances, the proposed continuous-time framework accommodates a broader range of delay processes and offers stronger stability guarantees. Lastly, it aligns well with the principles of input-to-state stability (ISS) and noise-to-state stability (NSS) in stochastic control systems, demonstrating that A-SGD converges to a noise-dependent neighborhood of the optimal parameter at an exponential rate despite the presence of delays. We verify our findings, in particular, on an example scenario of an over-parameterized neural network.[4]

Auxiliary Implications: Over-parameterized neural networks often satisfy a PL-like condition [17], leading to more rapid convergence [24], and sometimes global convergence [2, 3]. Our framework remains valid when delays vanish, reproducing standard SGD analysis. This connects the known benefits of over-parameterization with A-SGD by linking delay effects to faster convergence under PL-like assumptions. In [1], the study of over-parameterized regimes is extended to multi-layer neural networks, showing that for sufficiently wide architectures, stochastic gradient descent can reach global minima of deep training objectives in polynomial time. The authors demonstrate that wide networks exhibit dynamics closely approximated by neural tangent kernels, leading to linear convergence of SGD under mild assumptions. This result complements the Polyak–Lojasiewicz framework by establishing that large network width can inherently induce PL-like optimization landscapes, even for highly non-linear deep models.

[4] Our code can be found at https://github.com/arijitcodespace/Asynchronous-SGD.

4. Mathematical Formulation In this work, we analyze the parameter update for each individual worker as a continuous-time SDE of the form:

$$d\theta(t) = -\eta(t)\nabla f(\theta(t-\tau(t)))dt + \eta(t)\sigma(\theta(t-\tau(t)))dW(t), \tag{1}$$

where $f : \mathbb{R}^k \to \mathbb{R}$ is a real-valued differentiable function, $\eta(\cdot)$ is a learning rate to be specified later, $\theta(t) \in \mathbb{R}^k$, $\sigma^2(\theta(t))$ is the noise variance, $0 \le \tau(t) \le T < \infty$ is the delay and W is a standard k-dimensional Brownian motion. Let us first explain the meaning of the learning rate which we have used in the above equation.

REMARK 1. (**Learning rate**): Suppose we have the stochastic differential equation:

$$dx = \eta\left(f(x)\,dt + g(x)\,dW(t)\right),$$

where η is a learning rate taken to be constant, and $W(t)$ is a standard Brownian motion. We define a new time variable t' such that

$$t' = \eta t \quad \Rightarrow \quad dt = \frac{dt'}{\eta},$$

and a new Brownian motion $\widetilde{W}(t')$ by

$$\widetilde{W}(t') = \sqrt{\eta}\,W(t),$$

so that

$$dW(t) = \frac{d\widetilde{W}(t')}{\sqrt{\eta}},$$

thanks to the scaling property of the Brownian motion. Substituting into the original equation:

$$\begin{aligned} dx &= \eta\left(f(x)\,\frac{dt'}{\eta} + g(x)\,\frac{d\widetilde{W}(t')}{\sqrt{\eta}}\right) \\ &= f(x)\,dt' + \sqrt{\eta}\,g(x)\,d\widetilde{W}(t'). \end{aligned}$$

The original SDE

$$dx = \eta\left(f(x)\,dt + g(x)\,dW(t)\right)$$

can be rewritten, after a change of time and Brownian motion, as

$$dx = f(x)\,dt' + \sqrt{\eta}\,g(x)\,d\widetilde{W}(t'),$$

which shows that η acts as a rescaling of time and that the effective noise amplitude scales as $\sqrt{\eta}$.•

Throughout, for simplicity, we assume that $\sigma(\theta(t)) \equiv \sigma$ is a constant scalar; this can easily be generalized[5]. We therefore have that

$$d\theta(t) = -\eta(t)\nabla f(\theta(t-\tau(t)))dt + \eta(t)\sigma dW(t). \tag{2}$$

The main objective of this work is to analyze the convergence properties of (2).

5. Convergence for Polyak–Łojasiewicz Functions Throughout, we let θ^* denote a global minimizer of f, so that $f(\theta^*) = \inf_{\theta\in\mathbb{R}^k} f(\theta)$, and we abbreviate

$$V(t) := f(\theta(t)) - f(\theta^*) \geq 0, \qquad g_t := \nabla f(\theta(t)). \tag{3}$$

We define a random variable along (2), which we term *burn-in time*, as:

$$\tau_M := \inf\left\{t \geq 2T : \sup_{u\in[t-2T,\,t]} V(u) \leq M\right\}, \tag{4}$$

where the level $M > 0$ is the one fixed in Assumption 4. Requiring the whole delay window, rather than the single time t, to have entered the sublevel set costs nothing in the analysis and is what makes the delayed-history hypothesis of Lemma 6 available at time τ_M; note that τ_M is a stopping time, since V is continuous and adapted. Throughout we take $f \in C^2$, which is what allows Itô's Lemma to be applied to $f(\theta(\cdot))$. We now state the assumptions that we use throughout.

ASSUMPTION 1. (**Polyak–Lojasiewicz condition**): *There exists $\mu > 0$ such that for all $\theta \in \mathbb{R}^k$,*

$$\frac{1}{2}\|\nabla f(\theta)\|^2 \geq \mu\left(f(\theta) - f(\theta^*)\right). \tag{5}$$

ASSUMPTION 2. (**Lipschitz gradients**): *The gradient ∇f is L-Lipschitz continuous:*

$$\|\nabla f(\theta_1) - \nabla f(\theta_2)\| \leq L\|\theta_1 - \theta_2\|, \quad \forall \theta_1, \theta_2 \in \mathbb{R}^k. \tag{6}$$

ASSUMPTION 3. (**Bounded delay**): *The delay τ satisfies $\tau(t) \leq T < \infty$ for all $t \geq 0$. Throughout, $\tau : [0,\infty) \to [0,T]$ is a deterministic measurable function of time; Remark 11 explains how the analysis extends to random delays independent of the driving Brownian motion.*

Note that Assumption 3 is realistic, and although not uniformly assumed, for instance in [29], it has been widely used in the literature, e.g., [15].

[5] For example, with a matrix diffusion coefficient $\sigma \in \mathbb{R}^{k\times k}$, we replace $k\sigma^2$ by $\mathrm{tr}(\sigma\sigma^\top)$.

ASSUMPTION 4. (**Local gradient bound**): *There exist* $M, G_M \in \mathbb{R}$ *such that* $f(\theta) - f(\theta^*) \leq M \Rightarrow \|\nabla f(\theta)\| \leq G_M$, *where*

$$G_M = \sup_{\theta: f(\theta)-f(\theta^*)\leq M} \|\nabla f(\theta)\| < \infty.$$

The finiteness of G_M *is justified in the following assumption. This implication holds deterministically for all* $\theta \in \mathbb{R}^k$.

ASSUMPTION 5. (**Bounded sublevel sets**): *For all* $R < \infty$, $\Omega_R := \{\theta : f(\theta) - f(\theta^*) \leq R\}$ *is bounded. Thus, following from Assumption 2, we have that*

$$G_R := \sup_{\theta \in \Omega_R} \|\nabla f(\theta)\| < \infty.$$

ASSUMPTION 6. (**Learning rate**): *Given a delay bound* $T > 0$, *a final time* $0 < t_f < \infty$ *and a switching time* $t_0 \in [0, t_f)$, *we assume that the learning rate* $\eta : [0, t_f] \to \mathbb{R}_{>0}$ *satisfies*

1. (Continuity)*:* η *is continuous on* $[0, t_f]$*;*
2. (Delayed monotonicity)*:* η *is non-increasing on* $[t_0, t_f]$,

and that, for some $K \geq 1$, *there is a* $(K+1)$*-partition*

$$t_0 < t_1 < \cdots < t_K = t_f$$

of $[t_0, t_f]$ *satisfying* $t_{j+1} - t_j \geq T$ *for all* $j \in \{0, \ldots, K-1\}$. *We write*

$$\eta_+ := \max_{s\in[0,t_f]} \eta(s), \qquad \eta_- := \min_{s\in[0,t_f]} \eta(s),$$

which exist and are positive by continuity and compactness.

Note that we do not require any monotonicity property on the interval $[0, t_0)$; a partition of the required kind exists whenever $t_f - t_0 \geq T$, and K may be taken as large as $\lfloor (t_f - t_0)/T \rfloor$. The main result of this work is as follows:

THEOREM 1. (**Convergence of A-SGD under PL condition**): *Let Assumptions 1–5 hold, fix a horizon* $H > 0$ *and a failure level* $\delta \in (0, 1)$, *and suppose that the level* M *of Assumption 4 satisfies* $M \geq M_\star$ *with* $M_\star$ *as in Lemma 1. Assume* $\eta(t) = \eta$ *with* $\eta \in \mathbb{R}_{>0}$ *satisfying both the step-size condition (13) and* $\eta \leq \bar{\eta}$, *where* $\bar{\eta} = \bar{\eta}(M, H, \delta)$ *is as in Lemma 6. Then the trajectories of (2) satisfy, almost surely, for every* $t \in [\tau_M, \tau_M + H]$,

$$\mathbb{E}\left[V(t \wedge \rho) \,\middle|\, \mathcal{F}_{\tau_M}\right] \leq V(\tau_M)\, e^{-\eta\mu(t-\tau_M)} + (\kappa + 2M\delta)\left(1 - e^{-\eta\mu(t-\tau_M)}\right) + \eta^3 L^2 \sigma^2 T\, \Psi(\tau_M), \qquad (7)$$

where $\rho := \inf\{s \geq \tau_M : V(s) > 2M\}$ *satisfies* $\mathbb{P}(\rho \leq \tau_M + H \mid \mathcal{F}_{\tau_M}) \leq \delta$ *almost surely, where* $V(\tau_M) \leq M$*, where* Ψ *is the* $\mathcal{F}_{\tau_M}$*-measurable boundary term* (17)*, and where*

$$\kappa = \frac{\eta^2}{\mu}\left(L^2T^2G_{2M}^2 + kL^2\sigma^2T\right) + \frac{\eta\sigma^2kL}{2\mu}$$

is called the noise margin.

Three features of (7) deserve comment. First, the localization is genuinely needed: Ω_{2M} is not forward invariant for (2), so a bound valid for all t and all sample paths simultaneously is not available. The bound is a statement about the process stopped on exit from Ω_{2M}, and it controls $\mathbb{E}[V(t)\mathbb{1}_{\{\rho>t\}} \mid \mathcal{F}_{\tau_M}]$, with $\mathbb{P}(\rho > t \mid \mathcal{F}_{\tau_M}) \geq 1 - \delta$. Second, H and δ must be fixed before η, since $\bar{\eta}$ depends on both; δ then enters the margin only through the additive term $2M\delta$. Note that the $\bar{\eta}$ constructed in Lemma 6 decays like $1/H$, so the admissible horizon and step size are coupled; sharpening that dependence to a logarithmic one would require replacing the union-bound exit estimate of Lemma 6 by an exponential-supermartingale estimate. Third, the term $\eta^3L^2\sigma^2T\,\Psi(\tau_M)$ is the conditioning-boundary contribution discussed in Remark 10: it is history data observable at time τ_M, it acts only on a window of length T, and it is $O(\eta^3)$, hence of strictly higher order than the $O(\eta)$ leading term of κ. It is not bounded uniformly in the initialization; (24) bounds its conditional mean at deterministic times, but not at the stopping time τ_M.

In practice, e.g., in neural networks, one often employs a time-varying learning rate. In what follows, we provide an upper bound for A-SGD in such settings.

THEOREM 2. (**Convergence with time-varying** η): *Let* $t_f > 0$*, fix a failure level* $\delta \in (0,1)$*, and suppose that Assumptions 1–5 hold, that* $M \geq M_\star$ *with* $M_\star$ *as in Corollary 1, and that the learning rate* η *satisfies Assumption 6, the two-sided step-size condition* (34)*, and* $\eta_+ \leq \bar{\eta}$ *with* $\bar{\eta} = \bar{\eta}(M, t_f, \delta)$ *as in Lemma 6. Let* ρ *be as in Theorem 1. Then, almost surely on the event* $\{\tau_M \leq t_f\}$*, the parameter trajectory governed by the SDE* (2) *satisfies the convergence bound*

$$\mathbb{E}\left[V(t\wedge\rho)\,\middle|\,\mathcal{F}_{\tau_M}\right] \leq V(\tau_M)\,e^{-\mu\int_{\tau_M}^{t}\eta(s)ds} + \kappa(K) + 2M\delta + \eta_+L^2\sigma^2T\,\Psi_\eta(\tau_M) \tag{8}$$

for $t \in [\tau_M, t_f]$*, where, with the convention* $t_{-1} := 0$*,*

$$\kappa(K) := \sum_{j=-1}^{K-1}\frac{\beta_{max,j}}{\mu\eta_-}\left(1 - e^{-\mu\eta_-\left(t_{j+1}-t_j\right)}\right), \tag{9}$$

and

$$\beta_{max,j} := L^2\hat{\eta}_j^3\left(T^2G_{2M}^2 + k\sigma^2T\right) + \frac{\sigma^2}{2}H_{max}\,\hat{\eta}_j^2. \tag{10}$$

Here $\eta_{max,\,j}$ is the maximum of η on $[t_j, t_{j+1}]$, $\eta_\pm$ are as in Assumption 6,

$$\hat{\eta}_j := \sup_{u\in[(t_j-T)\vee 0,\ t_{j+1}]} \eta(u), \qquad H_{max} := \sup_{\theta\in\Omega_{2M}} \left|\mathrm{tr}\left(\nabla^2 f(\theta)\right)\right| \le kL,$$

Ω_{2M} is as in Assumption 5, and

$$\Psi_\eta(s_0) := \sup_{a\in[s_0-T,\,s_0]} \left\| \int_a^{s_0} \eta(u)\, dW(u) \right\|^2 \tag{11}$$

is the $\mathcal{F}_{s_0}$-measurable conditioning-boundary term, which reduces to $\eta^2\Psi(s_0)$ for a constant learning rate. By Assumption 6, $\hat{\eta}_j \le \eta_{max,\,j-1}$ for $j \ge 1$ and $\hat{\eta}_{-1}, \hat{\eta}_0 \le \eta_+$ may be used for the two initial blocks. Moreover $\mathbb{P}(\tau_M > t_f) \le T_{burn}/t_f$ by Corollary 1 and Markov's inequality, and $\mathbb{P}(\rho \le t_f \mid \mathcal{F}_{\tau_M}) \le \delta$.

The block $j = -1$, namely $[0, t_0]$, carries the pre-decay phase, on which Assumption 6 imposes no monotonicity; the quantity $\hat{\eta}_j$ is needed because the delay window of a time in block j reaches back into block $j-1$, where η is larger. For a constant schedule $\hat{\eta}_j = \eta_{\max,\,j} = \eta_\pm = \eta$, all the $\beta_{\max,\,j}$ coincide, and keeping the discount factors instead of the uniform bound used in (9) makes the sum in the proof telescope to

$$\kappa\left(1 - e^{-\eta\mu(t-\tau_M)}\right) \le \kappa,$$

with κ the noise margin of Theorem 1; together with $H_{\max} \le kL$ this recovers Theorem 1 exactly, the additive terms $2M\delta$ and $\eta^3 L^2\sigma^2 T\,\Psi(\tau_M)$ being common to both statements. The result has several other implications, which we discuss below.

REMARK 2. (**Exponential descent under PL**): Once the learning rate decays to a small value, the dominant term in both (7) and (8) is

$$V(\tau_M)\,\exp\!\left(-\mu\int_{\tau_M}^{t}\eta(s)\,ds\right).$$

Therefore, for objective functions which satisfy the PL condition, the trajectories of A-SGD contract (with high probability) at the same exponential rate as those of synchronous SGD. The effect of the delay is to reduce the admissible upper bound on η and to enlarge the noise margin; it does not change the *form* of the contraction. •

REMARK 3. (**Noise margin versus step-size tail**): The residual term in the proof (see Appendix D) is the convolution of the forcing β with the exponential discount $e^{-\mu\int_u^t \eta(s)ds}$. It is this discount, and not the behaviour of the accumulated Brownian variance $\int_{\tau_M}^t \eta^2(s)ds$, that makes

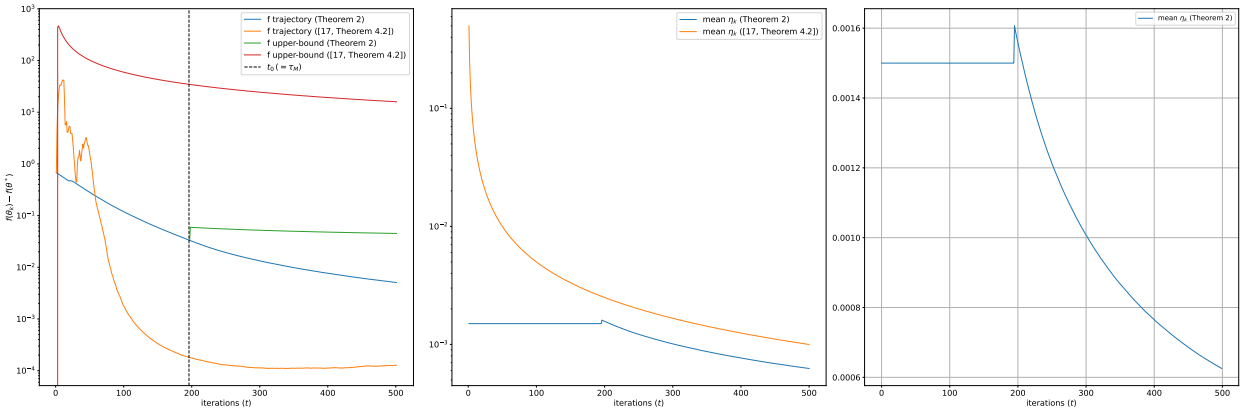


FIGURE 2. Experiment 1: Trajectories for A-SGD and corresponding upper bounds for Theorem 2 and [10, Theorem 4.2] (Left); learning rate schedules in log-scale (Middle); learning rate for Theorem 2 (linear scale) (Right).

the residual stay bounded; for *constant* or *block-wise constant* schedules that accumulated variance grows without bound as $t_f \to \infty$, yet the residual still stabilizes at the strictly positive value κ of Theorem 1, which is the sense in which κ is a floor rather than a transient. •

REMARK 4. (**Practical speed-accuracy trade-off**): We observe from Theorem 2 that the learning rate only needs to decrease after the switching time t_0 of Assumption 6. The interval $[0, t_0]$ is the pre-decay phase, on which no monotonicity is imposed and during which the value gap $V(t)$ typically decays rapidly. In practice, one can set a relatively large learning rate in this phase so that the gap closes quickly (in fewer iterations) before the decay schedule takes effect. This drives the value gap rapidly to $O(\eta\sigma^2 kL/\mu)$, the dominant term of κ for small η. The subsequent decay of the learning rate then gradually shrinks this floor, albeit at the cost of slower convergence in the late stage. The two-sided condition (34) limits how aggressive this trade-off can be, since it couples η_+ to η_-. •

REMARK 5. (**Delay robustness**): The delay-dependent part of $\beta_{\max,j}$ in (10) reads

$$L^2\hat{\eta}_j^3\left(T^2G_{2M}^2 + k\sigma^2 T\right).$$

Once the learning rate begins to decay, this term contributes less to the overall value of $\kappa(K)$, and it does so at the third order in $\hat{\eta}_j$, faster than the delay-free term. Consequently, moderate asynchronous staleness affects only the constants in the analysis, without altering the qualitative behavior described above. •

6. Experiments and Results In this section, we present two kinds of experiments, one with a synthetic objective function and one with over-parameterized neural networks. We start

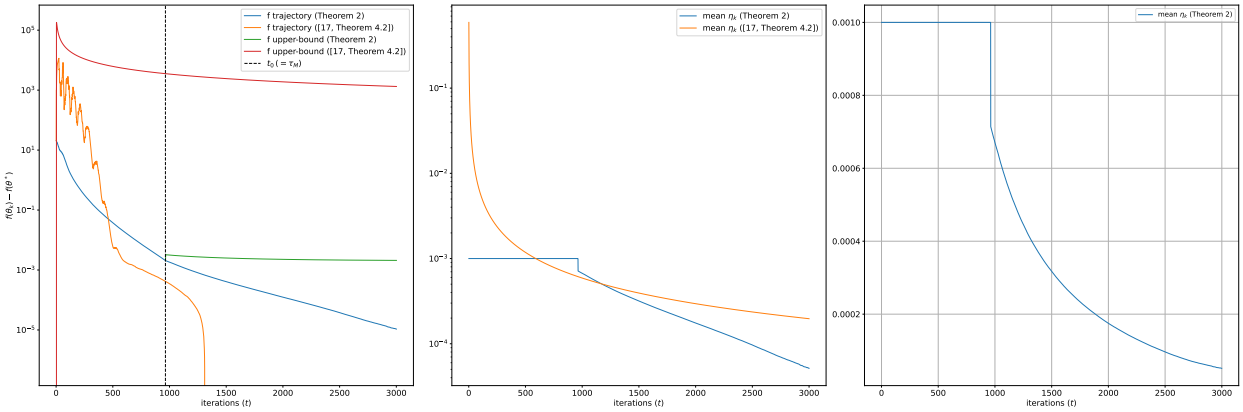


FIGURE 3. Experiment 2: Trajectories for A-SGD and corresponding upper bounds for Theorem 2 and [10, Theorem 4.2] (Left); learning rate schedules, upper and lower bounds (Middle); learning rate for Theorem 2 (linear scale) (Right).

our experiments from the simpler case of the synthetic objective function. We denote by $\theta^{(t)}$ the t-th iterate of θ. For brevity, we omit the algorithms for A-SGD simulation, Hessian trace, and L-smoothness estimation for neural networks, which are fairly standard algorithms used in the literature and can be found in the repository linked above.

6.1 Synthetic Objective Function

We choose the objective function:

$$f(\theta) = \theta^\top Q\theta + \epsilon \sin(w^\top \theta),$$

for $Q \succ \mathbf{0}$, $\epsilon \geq 0$ and $w \neq \mathbf{0}$. This function is strongly convex, and hence satisfies the Polyak–Lojasiewicz condition, whenever $\epsilon < 2\lambda_{\min}(Q)/\|w\|^2$. In that regime one may take as strong-convexity modulus (equivalently, as PL constant) the value (see Appendix E for a derivation)

$$\mu := 2\lambda_{\min}(Q) - \epsilon\|w\|^2,$$

and as L-smoothness constant

$$L := 2\lambda_{\max}(Q) + \epsilon\|w\|^2,$$

where $\lambda_{\min}(Q)$ and $\lambda_{\max}(Q)$ are the smallest and largest eigenvalues of Q, respectively. Both are admissible constants rather than the exact extrema, which is all the analysis requires.

Setting: Throughout this section, we focus solely on a strongly convex objective and compare our bound to others relevant in the literature. We verify the upper bound proposed by Theorem 2 for a time-varying learning rate schedule. The results are plotted in Figure 2 and Figure 3. For [10,

Parameter	**Theorem 2**	**[10, Theorem 4.2]**
Number of parallel A-SGDs	100	100
Initial norm $\|\theta_0\|$	0.5	0.5
Dimension	50	50
$\lambda_{\max}(Q)$	5.0	5.0
$\lambda_{\min}(Q)$	1.0	1.0
ϵ	10^{-4}	10^{-4}
iterations (t_f)	500	500
Maximum Delay (iterations)	20	20
σ^2 or $\mathrm{tr}(\sigma\sigma^\top)$	10^{-8}	10^{-8}
fraction of initial value gap after which learning rate decay starts	0.05	–

TABLE 1. Comparison of Parameters in Theorem 2 and [10, Theorem 4.2], Experiment 1.

Theorem 4.2], to obtain an upper bound of $f(\theta^{(t)}) - f(\theta^*)$, we simply multiply their upper bound by a factor of $L/2$.

For simplicity of simulation, in the case of synthetic functions, we further upper bound $\kappa(K)$, given in Theorem 2, by upper bounding

$$H_{\max} = \sup_{\theta \in \Omega_{2M}} \left|\mathrm{tr}(\nabla^2 f(\theta))\right| \leq kL.$$

We do this to avoid calculating the maximum trace of the Hessian throughout the simulation.

Results: We perform two experiments, the parameters of which are given in Table 1 and Table 2, respectively. The results are shown in Figure 2 and Figure 3. It should be noted that the trajectories shown in the figures are empirical mean trajectories across a number of parallel simulations. For the second experiment, we chose a time-varying noise variance instead of a fixed variance, and to plot the upper bound, we used the mean variance across iterations. In both experiments the trajectory and the corresponding upper bound improve substantially on those obtained in [10, Theorem 4.2]. For the learning rate schedules (see Figure 2 and Figure 3), we keep the learning rate fixed during the initial phase and start the decay only when the difference between the current objective value and the optimum has fallen below the threshold given in Table 1 and Table 2.

Parameter	**Theorem 2**	**[10, Theorem 4.2]**
Number of parallel A-SGDs	200	200
Initial norm $\|\theta_0\|$	2.0	2.0
Dimension	100	100
$\lambda_{\max}(Q)$	10.0	10.0
$\lambda_{\min}(Q)$	1.0	1.0
ϵ	10^{-2}	10^{-2}
iterations (t_f)	3000	3000
Maximum Delay (iterations)	50	50
σ^2 or $\mathrm{tr}(\sigma\sigma^\top)$	See Remark 6	See Remark 6
fraction of initial value gap after which learning rate decay starts	5×10^{-4}	–

TABLE 2. Comparison of Parameters in Theorem 2 and [10, Theorem 4.2], Experiment 2.

REMARK 6. (**Variance schedule in Experiment 2**): For Experiment 2, we chose a variance schedule instead of a fixed variance, i.e., the variance is time-varying:

$$\sigma_i^2 \equiv \mathrm{tr}(\sigma_i \sigma_i^\top) = 10^{-6}\left(e^{-5\times 10^{-3} i} + 10^{-10}\right),$$

where $i \in \{0, 1, \ldots, t_f - 1\}$ denotes the index of the iteration. While calculating the upper bound, we used the empirical mean of the schedule defined above:

$$\bar{\sigma}^2 = \frac{10^{-6}}{t_f} \sum_{i=0}^{t_f-1} \left(e^{-5\times 10^{-3} i} + 10^{-10}\right).$$

•

6.2 "Over-parameterized" Neural Networks

In this section, we validate our upper bound given by Theorem 2 on over-parameterized neural networks. Roughly speaking, a neural network is called "over-parameterized" if the number of parameters is larger than the number of data points. From [17], we know that this class of neural networks tends to satisfy the PL condition on the loss landscape.

Setting: We use the CIFAR-100 [14] dataset for our experiments, which contains 50,000 training and 10,000 test RGB images, each of size $32 \times 32 \times 3$. We use VGG-16 [25] as our neural network. To estimate the gradient variance, we use the gradient of a 4096-batch as a reference when obtaining the estimated gradient with `batch_size` samples. During the burn-in time, we opt for a learning rate schedule with a warm-up that turned out to be empirically sound. The PL constant μ is estimated by using:

$$\mu = \min_{\theta} \frac{\|\nabla f(\theta)\|^2}{2f(\theta)},$$

since $f(\theta^*) = 0$. Equivalently, 2μ is the largest slope such that every point of the scatter plot of $\|\nabla f(\theta)\|^2$ (vertical axis) against $f(\theta)$ (horizontal axis) lies on or above the line through the origin of that slope. It must be noted that we train an auxiliary network (with the same architecture as in the experiments) while estimating the PL constant.

To estimate the smallest admissible L-smoothness constant, which for twice differentiable functions is the supremum over θ of the spectral norm of the Hessian, we use the power iteration method at each iteration during a proxy training and then take the maximum over all these values. We also estimate the trace of the Hessian using Hutchinson's method [11]. It must be noted that the L-smoothness constant, like the PL constant μ, does not depend on the kind of optimizer used. It only depends on the loss landscape, and hence, to get near the optimum, we used the Adam optimizer with momentum and estimated these parameters. However, the term $H_{\max}$ does depend on the region the trajectory visits and, consequently, on the choice of the optimizer itself. To estimate the trace empirically while avoiding the computational overhead of running two A-SGDs (one to estimate the trace and one for simulation), we do the trace calculation along with the calculation of μ and L. We note that this is an empirical estimate and should be interpreted accordingly.

Finally, to simulate A-SGD, we use momentum during training. This is because large networks like VGG-16 are known to settle/oscillate at poor local minima when trained without momentum-based optimizers [22, 27]. This is *not* in contradiction with our theory because momentum-based optimization leads to better minima and thus the upper bound given by Theorem 2 still holds with some slack. Now, to simulate Theorem 2 we partition $[t_0, t_f]$ into a $(K+1)$-partition $\{t_0, t_1, \ldots, t_K\}$ with $t_K = t_f$ and $t_{j+1} - t_j \geq T$, as in Assumption 6. We perform an experiment on this network, the parameters of which are provided in Table 3.

REMARK 7. (**Scope of the empirical validation**): Two hypotheses of Theorem 2 are not verified by the reported experiments, and the comparisons below should be read accordingly. First,

Parameter	**Theorem 2**
Dimension (k)	138M
Maximum Delay (iterations)	10
σ^2 or $\mathrm{tr}(\sigma\sigma^\top)$	10^{-8}
iterations (t_f)	25000
learning rate schedule	$\begin{cases} 2\times10^{-8}t + 10^{-4}, & t \le 10000 \\ 3\times10^{-4}\exp\left(-4\times10^{-5}(t-10000)\right), & t > 10000 \end{cases}$ $t \longrightarrow$ index of iteration; $t \in \{1, 2, \dots\}$
$H_{\max}$	26250
μ	0.1273
L	33
f^*	0
fraction of initial value gap after which burn-in stops	0.3
G_{2M}	4.14
batch size	1024
Apply Momentum	True

TABLE 3. Parameter Settings for Experiment 3.

the two-sided step-size condition (34) is not met by the schedule of Table 3: with $\eta_- = 10^{-4}$, $\eta_+ = 3\times10^{-4}$, $\mu = 0.1273$, $L = 33$ and $T = 10$ one has $\eta_+^3 = 2.7\times10^{-11}$ against $\eta_-\mu/(12L^3T^2) = 2.95\times10^{-13}$, a violation by a factor of roughly 91; condition (34) is sufficient rather than necessary, and the bound is observed to hold with slack in this regime. Second, in Experiments 1 and 2 the decay is triggered when the value gap first falls below a preset fraction of its initial value, which makes the switching time data-dependent, whereas Assumption 6 takes t_0 deterministic; Table 3 uses the deterministic switching iteration $t_0 = 10000$. Finally, the plotted curves display the noise margin $\kappa(K)$ of (9), with $H_{\max}$ replaced by kL in Experiments 1 and 2 and by the Hutchinson estimate of Table 3 here; a fully certified bound would additionally include the terms $2M\delta$ and $\eta_+L^2\sigma^2T\,\Psi_\eta(\tau_M)$, which depend on the chosen failure level and on the realized history. •

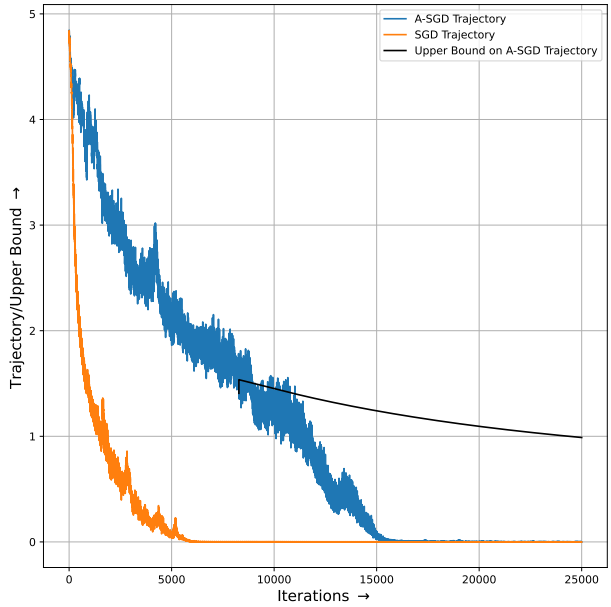


FIGURE 4. Experiment 3: Upper bound (Black), loss trajectory of A-SGD (Blue), and loss trajectory of SGD (Orange) of the neural network.

Results: Figure 4 shows the upper bound and loss trajectory under the settings of Table 3. The upper bound is valid after the process has reached a preset threshold (given in Table 3), i.e., after the trajectory has stayed inside Ω_M over a window of length $2T$, in the sense of (4). Since our bound is derived for vanilla A-SGD, intuitively and practically, it makes sense that we observed momentum-based A-SGD to obey the bound; this is because the loss trajectories for momentum-based techniques tend to remain lower than those without momentum (at least for convex objectives).

REMARK 8. (**Adaptive local bounds and a lower-envelope certificate**): Our analysis is local: it certifies the trajectory once it enters a chosen sublevel set Ω_M. This allows practitioners to tailor M or G_M to the phase of training, compute a small family of inexpensive certificates, and then take their point-wise lower envelope to produce a tight, global-in-time estimate. In contrast, fully non-local bounds (e.g., [10, Theorem 4.2]) are necessarily more conservative and may not capture the late-stage decay seen in practice. Operationally, one can pre-compute bounds at several checkpoints–from initialization up to the initial gradient norm–and use the lower envelope as a robust worst-case curve. See Figure 5. •

7. Conclusion and Future Work

In this work, we developed an approach to analyze the performance of A-SGD on the class of Polyak–Lojasiewicz functions and demonstrated its performance without the explicit need for delay compensation. Using an appropriate choice of

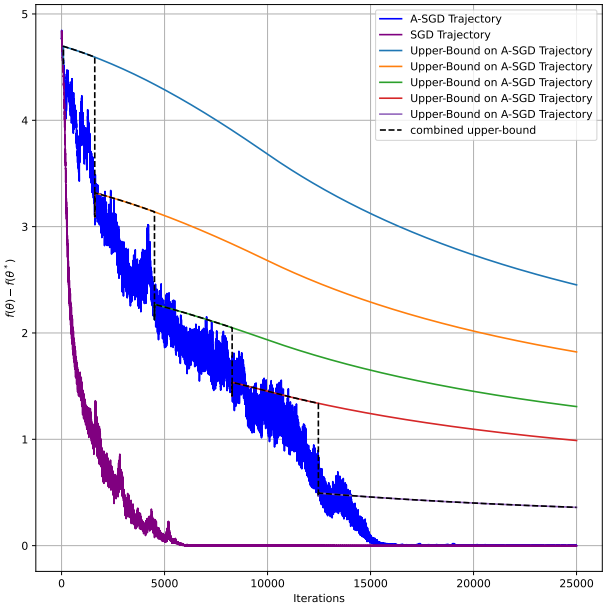


FIGURE 5. Minimum/combined upper bound (dashed) of the five calculated upper bounds (solid).

the learning rate schedule, we provided an upper bound for the underlying A-SGD process that is tighter than the previously known upper bound when restricted to the class of strongly convex functions. We demonstrated the effectiveness of the bound in a deep neural network experiment in an over-parameterized regime, whose loss landscape tends to satisfy a variant of the PL condition. Furthermore, we showed in an experiment that our bound can be applied to momentum-based optimization settings. Further refinements and tightening of the proposed bound require knowledge of the structure imposed on the loss landscape for deep neural networks. To this end, we close by outlining directions for extending the analysis to other, wider classes of functions–for example, the Kurdyka–Lojasiewicz class of functions.

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Appendix A: Bounded Burn-in Time We first state some lemmas that will be useful in proving Theorem 1 and Theorem 2. Recall the notation (3); throughout the appendices we adopt the history conventions $\theta(u) \equiv \theta(0)$ and $W(u) \equiv W(0)$ for $u \le 0$, so that $V(u) \equiv V(0)$ for $u \le 0$. In this notation, Assumption 1 reads $\frac{1}{2}\|g_t\|^2 \ge \mu V(t)$, and Assumption 2 gives both $\left\|g_t - g_{t-\tau(t)}\right\| \le L\|\theta(t) - \theta(t-\tau(t))\|$ and, since $f \in C^2$,

$$\operatorname{tr}\left(\nabla^2 f(\theta(t))\right) \le kL. \tag{12}$$

LEMMA 1. (**Bounded burn-in time**): *Under Assumptions 1–3, and for a constant learning rate η satisfying the step-size condition*

$$\eta\mu T \le 1 \quad \textit{and} \quad \eta^2 \le \frac{\mu}{12L^3T^2}, \tag{13}$$

there exists a deterministic constant $M_\star$ such that for every $M \ge M_\star$,

$$\mathbb{E}[\tau_M] \le T_{burn} < \infty.$$

Here $T > 0$; the delay-free case $T = 0$ reduces to the classical SGD setting. Some smallness requirement of the type (13) is unavoidable: already the noiseless delayed linear flow $\dot{x} = -\eta L\, x(t - T)$ is unstable when $\eta LT > \pi/2$, in which case $\mathbb{E}[\tau_M] = \infty$ can occur. Condition (13) is what we

mean by "a constant learning rate η" in the statement, and it is the quantitative counterpart of the bound $\eta \le \bar{\eta}$ produced by Lemma 6. We collect the constants used below:

$$\begin{aligned}
\varepsilon &:= \frac{\eta\mu}{2}, \qquad b_\eta := 2\eta^3 L^3 T^2, \\
c_\eta &:= \eta^3 L^2 \sigma^2 kT + \frac{1}{2}\eta^2 \sigma^2 kL, \\
\bar{c} &:= \frac{2\eta^2 L^2 \sigma^2 kT + \eta\sigma^2 kL}{\mu}, \\
\gamma_0 &:= 2 + 4\eta LT, \qquad \gamma_1 := (k+4)\eta^2 \sigma^2 LT, \\
t_\star &:= 5T + \frac{2}{\eta\mu}\log(8\gamma_0), \\
\Gamma &:= 2\gamma_0 \bar{c} + 2\gamma_1 + 2Tc_\eta + 32\eta^3 L^2 \sigma^2 kT^2, \\
M_\star &:= 4\Gamma.
\end{aligned} \tag{14}$$

With these constants the proof below yields the explicit value

$$T_{\text{burn}} = 2t_\star \left(1 + \frac{2V(0) + 2Tc_\eta}{M}\right) \le t_\star \left(\frac{5}{2} + \frac{4V(0)}{M}\right). \tag{15}$$

We control the *conditional first moment* of the objective gap. The PL condition pushes $\mathbb{E}[V(t)]$ down at the exponential rate $\varepsilon = \eta\mu/2$, while the delay contributes only a small self-referential term, the noise a bounded inflow, and the conditioning boundary a known finite-window forcing; a Halanay-type delay inequality then shows that, from *any* history, the conditional mean contracts to a noise floor $\bar{c}$. We tile time into blocks of the fixed length $t_\star$, so that Markov's inequality makes each block an *attempt* at dropping below level M. Tracking jointly the survival probability and the size of the surviving trajectories' history, the failure mass halves from block to block; hence τ_M has a geometric tail and finite expectation. In particular the argument is not localized to a sublevel set, so neither an exit time nor Assumptions 4–5 are needed here, and we condition only at deterministic times, so no strong Markov property is invoked.

Now we begin with the auxiliary results.

LEMMA 2. (**Gradient upper envelope**): *Under Assumption 2,*

$$\|\nabla f(\theta)\|^2 \le 2L\,(f(\theta) - f(\theta^*)), \quad \forall\, \theta \in \mathbb{R}^k. \tag{16}$$

Proof. By the descent lemma for L-smooth functions,

$$f(\theta^*) \le f\left(\theta - \tfrac{1}{L}\nabla f(\theta)\right) \le f(\theta) - \frac{1}{2L}\|\nabla f(\theta)\|^2,$$

which rearranges to (16). □

REMARK 9. (**Integrability and localization**): Since $\|\nabla f(\theta)\| \le \|\nabla f(\theta^*)\| + L\|\theta - \theta^*\|$, the drift of (2) has linear growth and the solution does not explode. All expectations below are first derived for the process stopped at $\zeta_j := \inf\{t : \|\theta(t) - \theta^*\| \ge j\}$, $j \in \mathbb{N}$, for which the stochastic integrals are true martingales; the resulting bounds are uniform in j, and we let $j \to \infty$ using Fatou's lemma and monotone convergence. We suppress this routine localization from now on. A crude a priori estimate (Grönwall with delay, using $\eta \left| g_t^\top g_{t-\tau(t)} \right| \le \eta L \left(V(t) + V(t - \tau(t)) \right)$ via Lemma 2) shows $\sup_{[0,t]} \mathbb{E}[V] < \infty$ for every t, so all quantities below are finite. •

When we condition on $\mathcal{F}_{s_0}$, the delay window of a time $u \in [s_0, s_0 + T)$ may straddle s_0: for $a := u - \tau(u) < s_0$ the Brownian increment $W(u) - W(a)$ contains the $\mathcal{F}_{s_0}$-measurable – random and unbounded – piece $W(s_0) - W(a)$, so that $\mathbb{E}[\|W(u) - W(a)\|^2 \,|\, \mathcal{F}_{s_0}] \ne k\tau(u)$ in general. The correct statement keeps this boundary piece as part of the (known) history data. We therefore define the $\mathcal{F}_{s_0}$-measurable random variables

$$\begin{aligned} \Phi(s_0) &:= \sup_{u \in [s_0 - 2T,\, s_0]} V(u), \\ \Psi(s_0) &:= \sup_{a \in [s_0 - T,\, s_0]} \|W(s_0) - W(a)\|^2, \end{aligned} \tag{17}$$

so that $\Phi(0) = V(0)$ and $\Psi(0) = 0$ by the history conventions.

LEMMA 3. (**Conditional energy inequality**): *Fix $s_0 \ge 0$ and set $m(t) := \mathbb{E}[V(t) \,|\, \mathcal{F}_{s_0}]$ for $t \ge s_0$ and $m(t) := V(t)$ for $t \in [s_0 - 2T, s_0]$. Then, almost surely, for all $s_0 \le s \le t$,*

$$m(t) \le m(s) + \int_s^t \left[-\eta\mu\, m(u) + b_\eta\, \bar{m}(u) + c_\eta + \eta^3 L^2 \sigma^2 \Psi(s_0) \mathbb{1}_{\{u < s_0 + T\}} \right] du, \tag{18}$$

where $\bar{m}(u) := \sup_{w \in [u-2T,\, u]} m(w)$, and b_η, c_η are as in (14).

Proof. By Itô's Lemma [12] applied to $f(\theta(u))$, for $u \ge s_0$,

$$dV(u) = -\eta\, g_u^\top g_{u-\tau(u)}\, du + \frac{1}{2}\eta^2\sigma^2 \operatorname{tr}\left(\nabla^2 f(\theta(u)) \right) du + \eta\sigma\, g_u^\top dW(u). \tag{19}$$

Using $-g_u^\top g_{u-\tau(u)} = -\|g_u\|^2 + g_u^\top \left(g_u - g_{u-\tau(u)} \right)$, Young's inequality, and Assumption 2,

$$\begin{aligned} -\eta\, g_u^\top g_{u-\tau(u)} &\le -\frac{\eta}{2}\|g_u\|^2 + \frac{\eta}{2}\left\| g_u - g_{u-\tau(u)} \right\|^2 \\ &\le -\frac{\eta}{2}\|g_u\|^2 + \frac{\eta L^2}{2}\|\theta(u) - \theta(u - \tau(u))\|^2. \end{aligned} \tag{20}$$

For the displacement, write $a := u - \tau(u)$; by (2),

$$\theta(u) - \theta(a) = -\eta \int_a^u g_{w-\tau(w)} dw + \eta\sigma \left(W(u) - W(a)\right),$$

and $\|x+y\|^2 \le 2\|x\|^2 + 2\|y\|^2$ splits the two contributions. The drift part is handled pathwise by Cauchy–Schwarz and $\tau(u) \le T$; its conditional mean involves $\mathbb{E}[\left\|g_{w-\tau(w)}\right\|^2 \mid \mathcal{F}_{s_0}] \le 2L\, m(w-\tau(w))$ by Lemma 2 (for $w - \tau(w) \le s_0$ these are the known history values, and times ≤ 0 are covered by the history conventions), with $w - \tau(w) \in [u-2T, u]$, and is therefore at most $2\eta^2 T \cdot 2LT\, \bar{m}(u)$. For the Brownian part we distinguish two cases.

(i) $a \ge s_0$. Then $W(u) - W(a)$ consists of post-s_0 increments only, so that

$$\mathbb{E}\left[\|W(u) - W(a)\|^2 \,\middle|\, \mathcal{F}_{s_0}\right] = k\tau(u) \le kT.$$

(ii) $a < s_0$, which is possible only when $u < s_0 + T$, since $u - a \le T$. Splitting

$$W(u) - W(a) = \left(W(s_0) - W(a)\right) + \left(W(u) - W(s_0)\right),$$

the first summand is $\mathcal{F}_{s_0}$-measurable and the second is independent of $\mathcal{F}_{s_0}$ with conditional mean zero, so the cross term vanishes and

$$\mathbb{E}\left[\|W(u) - W(a)\|^2 \,\middle|\, \mathcal{F}_{s_0}\right] = \|W(s_0) - W(a)\|^2 + k(u - s_0) \le \Psi(s_0) + kT.$$

In both cases $\mathbb{E}[\|W(u) - W(a)\|^2 \mid \mathcal{F}_{s_0}] \le kT + \Psi(s_0)\mathbb{1}_{\{u < s_0+T\}}$, and therefore

$$\mathbb{E}\left[\|\theta(u) - \theta(u-\tau(u))\|^2 \,\middle|\, \mathcal{F}_{s_0}\right] \le 4\eta^2 LT^2\, \bar{m}(u) + 2\eta^2\sigma^2 kT + 2\eta^2\sigma^2 \Psi(s_0)\mathbb{1}_{\{u<s_0+T\}}. \tag{21}$$

Finally, Assumption 1 gives $\mathbb{E}[\|g_u\|^2 \mid \mathcal{F}_{s_0}] \ge 2\mu\, m(u)$. Taking conditional expectations in (19) – the stochastic integral has zero conditional mean by Remark 9 – and combining (12), (20) and (21) yields (18); note that $\frac{\eta L^2}{2} \cdot 2\eta^2\sigma^2 = \eta^3 L^2 \sigma^2$ and $\frac{\eta L^2}{2} \cdot 4\eta^2 LT^2 = b_\eta$. The resulting inequality holds almost surely simultaneously for all rational $s \le t$, hence for all $s \le t$ by continuity of $u \mapsto m(u)$.

□

LEMMA 4. (**Halanay inequality**): *Let $m : [-2T, \infty) \to [0, \infty)$ be continuous and satisfy, for all $0 \le s \le t$,*

$$m(t) \le m(s) + \int_s^t [-a\, m(u) + b\, \bar{m}(u) + c]\, du,$$

with constants $a > b \ge 0$, $c \ge 0$, and $\bar{m}$ as in Lemma 3. If $\varepsilon > 0$ satisfies $\varepsilon \le a - be^{2T\varepsilon}$, then

$$m(t) \le \frac{c}{a-b} + \Lambda e^{-\varepsilon t}, \qquad \Lambda := \sup_{[-2T,0]} m, \quad t \ge 0.$$

Proof. Since m and $\bar{m}$ are continuous, taking $s = t - h$ in the integral inequality and letting $h \downarrow 0$ gives, for every $t > 0$,

$$\limsup_{h \downarrow 0} \frac{m(t) - m(t-h)}{h} \le -a\, m(t) + b\, \bar{m}(t) + c. \tag{22}$$

Let $\bar{c} := c/(a-b)$, fix $\delta > 0$ and $\varepsilon' \in (0, \varepsilon)$, and set $\psi(t) := \bar{c} + \delta + (\Lambda + \delta)e^{-\varepsilon' t}$ for $t \ge 0$ and $\psi(t) := \psi(0)$ for $t \in [-2T, 0)$. Then $m < \psi$ on $[-2T, 0]$. Suppose $t_1 := \inf\{t \ge 0 : m(t) \ge \psi(t)\} < \infty$; by continuity $t_1 > 0$, $m(t_1) = \psi(t_1)$ and $m \le \psi$ on $[-2T, t_1]$. Since ψ is non-increasing, $\bar{m}(t_1) \le \bar{c} + \delta + (\Lambda + \delta)e^{-\varepsilon'(t_1 - 2T)}$, whence

$$\begin{aligned} -a\, m(t_1) + b\, \bar{m}(t_1) + c &\le -(a-b)\delta + (\Lambda + \delta)e^{-\varepsilon' t_1}\left(-a + be^{2T\varepsilon'}\right) \\ &< -\varepsilon'(\Lambda + \delta)e^{-\varepsilon' t_1} = \psi'(t_1), \end{aligned}$$

where we used $(a-b)\bar{c} = c$, $-a + be^{2T\varepsilon'} \le -a + be^{2T\varepsilon} \le -\varepsilon \le -\varepsilon'$, and $\delta > 0$. On the other hand, $m < \psi$ on $[0, t_1)$ together with $m(t_1) = \psi(t_1)$ gives

$$\frac{m(t_1) - m(t_1 - h)}{h} > \frac{\psi(t_1) - \psi(t_1 - h)}{h}$$

for all small $h > 0$, so the left-hand side of (22) at $t = t_1$ is at least $\psi'(t_1)$ – contradicting the display above. Hence $m < \psi$ on $[0, \infty)$; letting $\delta \downarrow 0$ and $\varepsilon' \uparrow \varepsilon$ gives the claim. □

LEMMA 5. (**Window supremum**): *Fix $0 \le s_0 \le r$ and let m be as in Lemma 3. Then, almost surely,*

$$\mathbb{E}\left[\sup_{t \in [r, r+2T]} V(t) \,\middle|\, \mathcal{F}_{s_0}\right] \le \gamma_0 \overline{M} + \gamma_1, \tag{23}$$

where $\overline{M} := \sup_{u \in [r-2T, r+2T]} m(u)$ and γ_0, γ_1 are as in (14). Moreover,

$$\mathbb{E}\left[\Psi(r+T) \,\middle|\, \mathcal{F}_{s_0}\right] \le 16kT. \tag{24}$$

Proof. For $t \in [r, r+2T]$, integrating (19) gives

$$V(t) = V(r) + \int_r^t \left(-\eta g_u^\top g_{u-\tau(u)} + \tfrac{1}{2}\eta^2\sigma^2 \operatorname{tr} \nabla^2 f(\theta(u))\right) du + N(t),$$

where $N(t) := \int_r^t \eta\sigma\, g_u^\top dW(u)$. Bounding the inner product by $\frac{1}{2}(\|g_u\|^2 + \|g_{u-\tau(u)}\|^2)$ and using (12),

$$\sup_{[r,r+2T]} V \le V(r) + \int_r^{r+2T} \frac{\eta}{2}\left(\|g_u\|^2 + \|g_{u-\tau(u)}\|^2\right) du + \eta^2\sigma^2 kLT + \sup_{[r,r+2T]} |N(t)|.$$

We now take $\mathbb{E}[\,\cdot\,|\,\mathcal{F}_{s_0}]$. Both u and $u-\tau(u)$ lie in $[r-2T, r+2T]$, so by Lemma 2 the integral term is at most $\frac{\eta}{2}\cdot 2L\cdot 2\overline{M}\cdot 2T = 4\eta LT\,\overline{M}$. By the conditional Doob L^2-inequality,

$$\begin{aligned}\mathbb{E}\left[\sup_{[r,r+2T]} |N(t)| \,\middle|\, \mathcal{F}_{s_0}\right]^2 &\le \mathbb{E}\left[\sup_{[r,r+2T]} N(t)^2 \,\middle|\, \mathcal{F}_{s_0}\right] \\ &\le 4\eta^2\sigma^2 \int_r^{r+2T} \mathbb{E}\left[\|g_u\|^2 \,\middle|\, \mathcal{F}_{s_0}\right] du \le 16\eta^2\sigma^2 LT\,\overline{M},\end{aligned}$$

so that, by $2\sqrt{xy} \le x + y$,

$$\mathbb{E}\left[\sup_{[r,r+2T]} |N(t)| \,\middle|\, \mathcal{F}_{s_0}\right] \le 4\eta\sigma\sqrt{LT\overline{M}} \le \overline{M} + 4\eta^2\sigma^2 LT.$$

Collecting terms and using $\mathbb{E}[V(r)\,|\,\mathcal{F}_{s_0}] = m(r) \le \overline{M}$ gives (23) with $\gamma_0 = 2 + 4\eta LT$ and $\gamma_1 = (k+4)\eta^2\sigma^2 LT$.

For (24), all increments defining $\Psi(r+T)$ lie in $[r, r+T] \subseteq [s_0, \infty)$, so with $\widetilde{W}(t) := W(r+t) - W(r)$, $t \in [0,T]$, a Brownian motion independent of $\mathcal{F}_{s_0}$,

$$\Psi(r+T) = \sup_{t\in[0,T]} \left\|\widetilde{W}(T) - \widetilde{W}(t)\right\|^2 \le 4 \sup_{t\in[0,T]} \left\|\widetilde{W}(t)\right\|^2,$$

and Doob's L^2-inequality, applied to the non-negative submartingale $\left\|\widetilde{W}(t)\right\|$, gives $\mathbb{E}[\sup_{[0,T]}\left\|\widetilde{W}(t)\right\|^2] \le 4\mathbb{E}\left\|\widetilde{W}(T)\right\|^2 = 4kT$. □

We can now prove Lemma 1.

Proof of Lemma 1 **Conditional Contraction:** We first check that $\varepsilon = \eta\mu/2$ is admissible in Lemma 4 for $a = \eta\mu$, $b = b_\eta$ and $c = c_\eta$. By (13),

$$b_\eta = 2\eta L^3 T^2 \cdot \eta^2 \le 2\eta L^3 T^2 \cdot \frac{\mu}{12 L^3 T^2} = \frac{\eta\mu}{6}, \tag{25}$$

and $e^{2T\varepsilon} = e^{\eta\mu T} \le e$, hence $b_\eta e^{2T\varepsilon} \le \frac{e}{6}\eta\mu \le \frac{\eta\mu}{2}$ and $a - b_\eta e^{2T\varepsilon} \ge \eta\mu/2 = \varepsilon$. Moreover $a - b_\eta \ge \frac{5}{6}\eta\mu$, so that

$$\frac{c_\eta}{a - b_\eta} \le \frac{2c_\eta}{\eta\mu} = \frac{2\eta^2 L^2 \sigma^2 kT + \eta\sigma^2 kL}{\mu} = \bar{c}, \tag{26}$$

and $b_\eta T \le \frac{\eta\mu T}{6} \le \frac{1}{6}$. Fix $s_0 \ge 0$ and write

$$c' := c_\eta + \eta^3 L^2 \sigma^2 \Psi(s_0).$$

Boundary layer $[s_0, s_0 + T]$. Dropping $-\eta\mu\, m \le 0$ in (18) and setting $h(u) := \sup_{[s_0 - 2T,\, u]} m$, we get $\bar{m} \le h$ and hence

$$h(u) \le \Phi(s_0) + \int_{s_0}^{u} \left(b_\eta h(w) + c'\right) dw, \quad u \in [s_0, s_0 + T],$$

since the integrand is non-negative and h is non-decreasing. By Grönwall's inequality and $b_\eta T \le \frac{1}{6}$,

$$\sup_{[s_0,\, s_0 + T]} m \le (\Phi(s_0) + c'T)\, e^{b_\eta T} \le 2\Phi(s_0) + 2c'T =: \widetilde{\Phi}(s_0). \tag{27}$$

Halanay from $s_0 + T$. For $u \ge s_0 + T$ the indicator in (18) vanishes, since then $u - \tau(u) \ge s_0$. Thus Lemma 4 applies with time origin $s_0 + T$ and history window $[s_0 - T, s_0 + T]$, on which $m \le \max\{\Phi(s_0), \widetilde{\Phi}(s_0)\} = \widetilde{\Phi}(s_0)$. Hence for $t \ge T$ – and trivially for $t \in [0, T]$, where $e^{-\varepsilon(t-T)} \ge 1 -$

$$\mathbb{E}\left[V(s_0 + t) \,\middle|\, \mathcal{F}_{s_0}\right] \le \bar{c} + \widetilde{\Phi}(s_0) e^{-\varepsilon(t-T)}, \qquad t \ge 0, \tag{28}$$

where, by (27),

$$\widetilde{\Phi}(s_0) = 2\Phi(s_0) + 2T\left(c_\eta + \eta^3 L^2 \sigma^2 \Psi(s_0)\right) \tag{29}$$

is $\mathcal{F}_{s_0}$-measurable, and $\widetilde{\Phi}(0) = 2V(0) + 2Tc_\eta$.

Block Recursion: Let $S_n := nt_\star$ with $t_\star \ge 5T$ as in (14), and define

$$u_n := \mathbb{P}(\tau_M > S_n), \qquad v_n := \mathbb{E}\left[\mathbb{1}_{\{\tau_M > S_n\}} \widetilde{\Phi}(S_n)\right],$$

so that $u_0 = 1$ and $v_0 = 2V(0) + 2Tc_\eta$.

A window estimate. Since $\Phi(S_{n+1}) = \sup_{[S_{n+1}-2T,\, S_{n+1}]} V$, Lemma 5 with $s_0 = S_n$ and $r = S_{n+1} - 2T$ (which satisfies $r \geq S_n$, as $t_\star \geq 2T$) gives

$$\begin{aligned}\mathbb{E}\left[\Phi(S_{n+1}) \,\middle|\, \mathcal{F}_{S_n}\right] &\leq \gamma_0 \sup_{u \in [S_{n+1}-4T,\, S_{n+1}]} \mathbb{E}\left[V(u) \,\middle|\, \mathcal{F}_{S_n}\right] + \gamma_1 \\ &\leq \gamma_0 \left(\bar{c} + \widetilde{\Phi}(S_n) e^{-\varepsilon(t_\star - 5T)}\right) + \gamma_1, \end{aligned} \tag{30}$$

where the second inequality is (28): on the whole window $u - S_n \geq t_\star - 4T \geq T$, so the exponent there is at most $-\varepsilon(t_\star - 5T)$.

Bound on u_{n+1}. By (4), on $\{\tau_M > S_{n+1}\}$ we have $\Phi(S_{n+1}) > M$ (note $S_{n+1} \geq t_\star \geq 2T$), and $\mathbb{1}_{\{\tau_M > S_n\}}$ is $\mathcal{F}_{S_n}$-measurable; hence by the conditional Markov inequality and (30),

$$\begin{aligned} u_{n+1} &\leq \mathbb{E}\left[\mathbb{1}_{\{\tau_M > S_n\}} \mathbb{P}\left(\Phi(S_{n+1}) > M \,\middle|\, \mathcal{F}_{S_n}\right)\right] \\ &\leq \frac{1}{M} \mathbb{E}\left[\mathbb{1}_{\{\tau_M > S_n\}} \left(\gamma_0 \bar{c} + \gamma_1 + \gamma_0 e^{-\varepsilon(t_\star - 5T)} \widetilde{\Phi}(S_n)\right)\right] \\ &= \frac{(\gamma_0 \bar{c} + \gamma_1)\, u_n + \gamma_0 e^{-\varepsilon(t_\star - 5T)} v_n}{M}. \end{aligned} \tag{31}$$

Bound on v_{n+1}. Since $\mathbb{1}_{\{\tau_M > S_{n+1}\}} \leq \mathbb{1}_{\{\tau_M > S_n\}}$, it suffices to bound the two random ingredients of $\widetilde{\Phi}(S_{n+1})$ in (29) conditionally on $\mathcal{F}_{S_n}$. The first is (30); for the second, by (24) with $r = S_{n+1} - T \geq S_n$ we have $\mathbb{E}[\Psi(S_{n+1}) \,|\, \mathcal{F}_{S_n}] \leq 16kT$. Therefore

$$v_{n+1} \leq 2\gamma_0 e^{-\varepsilon(t_\star - 5T)} v_n + \Gamma u_n, \tag{32}$$

with Γ as in (14).

Contraction of the Pair: By the choice of $t_\star$ in (14) we have $\varepsilon(t_\star - 5T) = \log(8\gamma_0)$, so that

$$\gamma_0 e^{-\varepsilon(t_\star - 5T)} = \frac{1}{8}, \qquad 2\gamma_0 e^{-\varepsilon(t_\star - 5T)} = \frac{1}{4}.$$

For $M \geq M_\star = 4\Gamma$ we have $\Gamma/M \leq \frac{1}{4}$ and, since $\Gamma \geq 2(\gamma_0 \bar{c} + \gamma_1)$, also $(\gamma_0 \bar{c} + \gamma_1)/M \leq \frac{1}{8}$. Setting $w_n := u_n + v_n/M$, (31) and (32) give

$$u_{n+1} \leq \frac{1}{8} u_n + \frac{1}{8} \frac{v_n}{M}, \qquad \frac{v_{n+1}}{M} \leq \frac{1}{4} \frac{v_n}{M} + \frac{1}{4} u_n,$$

and hence $w_{n+1} \le \frac{3}{8} w_n \le \frac{1}{2} w_n$. Since $w_0 = 1 + (2V(0) + 2Tc_\eta)/M$, induction yields

$$\mathbb{P}(\tau_M > S_n) = u_n \le w_n \le 2^{-n} w_0. \tag{33}$$

Finite Burn-In Time: As $t \mapsto \mathbb{P}(\tau_M > t)$ is non-increasing, (33) gives

$$\mathbb{E}[\tau_M] = \int_0^\infty \mathbb{P}(\tau_M > t)dt \le t_\star \sum_{n\ge 0} \mathbb{P}(\tau_M > S_n) \le t_\star w_0 \sum_{n\ge 0} 2^{-n} = 2t_\star \left(1 + \frac{2V(0) + 2Tc_\eta}{M}\right),$$

which is the value of T_{burn} announced in (15); the simplified bound there follows from $2Tc_\eta \le \Gamma \le M/4$. □

REMARK 10. (**Conditioning boundary**): The term $\Psi(s_0)$ in Lemma 3 is forced by the delay: for $u \in [s_0, s_0 + T)$ the delay window straddles the conditioning time, and the piece $W(s_0) - W(u - \tau(u))$ of the Brownian increment is $\mathcal{F}_{s_0}$-measurable, random and of unbounded support, so it cannot be averaged away. It is handled by keeping it as history data: it acts only on a layer of length T, and at the next block boundary its conditional mean is at most the deterministic constant $16kT$ by (24), because the window defining $\Psi(S_{n+1})$ contains only post-S_n increments. It is never needed pathwise and never exponentiated, which is why it does not destabilize the recursion (32). •

REMARK 11. (**Random delays**): Assumption 3 treats τ as a deterministic measurable function of time, which is what makes $W(s_0) - W(u - \tau(u))$ an $\mathcal{F}_{s_0}$-measurable quantity in case *(ii)* of the proof of Lemma 3. If instead τ is a random delay path independent of W and of $\theta(0)$, the argument applies verbatim under $\mathbb{P}(\cdot \mid \tau)$ with the filtration $\mathcal{G}_t := \mathcal{F}_t \vee \sigma(\tau)$, for which W remains a Brownian motion. All the constants in (14) depend on τ only through the bound T, so the resulting estimate on $\mathbb{E}[\tau_M \mid \tau]$ is uniform over $\{\tau : 0 \le \tau \le T\}$ and the conclusion follows by taking expectations. •

REMARK 12. (**Order of the constants**): As $\eta \to 0$ we have $\Gamma = 2\gamma_0 \bar{c}\,(1 + o(1))$, so that $M_\star$ is of the order of $\eta\sigma^2 kL/\mu$; that is, the burn-in level sits at the same order as the noise margin κ of Theorem 1, which contains the term $\eta\sigma^2 kL/(2\mu)$. Likewise $T_{\text{burn}} = O\left(\frac{1}{\eta\mu}\,(1 + V(0)/M)\right)$, the natural PL time-scale. Both constants are deterministic; in particular $M_\star$ depends neither on the initialization nor on M itself, so the requirement $M \ge M_\star$ is not circular. •

We note below that the result obtained in Lemma 1 holds, with an identical proof, even when the parameter η depends on time.

COROLLARY 1. (**Bounded burn-in time for time-varying** η): *Let* $\eta \equiv \eta(t)$, *with* $\eta : [0, t_f] \to \mathbb{R}_{>0}$ *satisfying* $\eta_- \le \eta(t) \le \eta_+$ *for all* $t \in [0, t_f]$, *extended to* $[0, \infty)$ *by* $\eta(t) := \eta(t_f)$ *for* $t > t_f$ *(which preserves* $\eta_\pm$ *and the condition below), where*

$$\eta_- \mu T \le 1 \quad \text{and} \quad \eta_+^3 \le \frac{\eta_- \mu}{12 L^3 T^2}. \tag{34}$$

Then $\mathbb{E}[\tau_M] \le T_{burn} < \infty$ *for every* $M \ge M_\star$, *with* T_{burn} *as in* (15) *and the constants formed as described below.*

Proof. The proof of Lemma 1 applies verbatim upon replacing η by η_+ in every term that is upper bounded, and by η_- in the contraction rate. Concretely, Lemma 3 holds with $a = \eta_- \mu$, $b = b_{\eta_+}$ and $c = c_{\eta_+}$, and (34) gives $b_{\eta_+} \le \eta_- \mu / 6$ exactly as in (25), so that $\varepsilon := \eta_- \mu / 2$ remains admissible in Lemma 4 and $b_{\eta_+} T \le 1/6$. The boundary term of Lemma 3 must be read as $\eta_+ L^2 \sigma^2 \Psi_\eta(s_0)$ with Ψ_η as in (11), since a time-weighted stochastic integral is not dominated pathwise by $\eta_+^2 \Psi(s_0)$; the estimate (24) is re-derived verbatim for Ψ_η, by Doob's inequality applied to $t \mapsto \int_r^{r+t} \eta(u) dW(u)$, and gives $\mathbb{E}[\Psi_\eta(r+T) \,|\, \mathcal{F}_{s_0}] \le 16 k \eta_+^2 T$, so that the last term of Γ is unchanged. The constants $b_\eta, c_\eta, \gamma_0, \gamma_1, \Gamma$ and $M_\star$ in (14) are formed with η_+, whereas

$$\bar{c} := \frac{2 c_{\eta_+}}{\eta_- \mu}, \qquad t_\star := 5T + \frac{2}{\eta_- \mu} \log(8 \gamma_0),$$

which is (26) with $a - b \ge \frac{5}{6} \eta_- \mu$. The constants $\eta_\pm$ of Assumption 6 exist by continuity and compactness; condition (34) is a genuine restriction on how far apart they may be. □

Appendix B: Staying in a Local Sublevel Set

LEMMA 6. (**Staying in a local sublevel set**): *Let Assumptions 2, 3 and 5 hold. Fix* $M > 0$, *a horizon* $H > 0$, *and a failure level* $\delta \in (0, 1)$, *and assume that, for some* $t_0 \ge T$, $f(\theta(t_0)) - f(\theta^*) \le M$ *and that the delayed history over the last* T *units of time already lies inside the slightly larger set*

$$\Omega_{2M} := \{\theta : f(\theta) - f(\theta^*) \le 2M\},$$

that is, $\theta(u) \in \Omega_{2M}$ *for all* $u \in [t_0 - T, t_0]$. *Then there exists* $\bar{\eta} > 0$, *depending on* $M, H, \delta, T, k, G_{2M}, L$ *and* σ, *such that for all constant learning rates* $\eta \in (0, \bar{\eta}]$,

$$\mathbb{P}\left(\sup_{t \in [t_0, t_0 + H]} V(t) > 2M \;\middle|\; \mathcal{F}_{t_0} \right) \le \delta$$

almost surely. If t_0 *is a stopping time, the same conclusion holds with* $\mathcal{F}_{t_0}$ *the stopped* σ*-algebra. For a time-varying learning rate the statement holds verbatim with* η *replaced by* η_+ *in the definition of* $\bar{\eta}$, *since* η *enters the proof only through upper bounds.*

Proof. Write $V(t) := f(\theta(t)) - f(\theta^*)$ and let

$$\varrho := \inf\{t \geq t_0 : V(t) > 2M\} \tag{35}$$

be the first exit from Ω_{2M} after t_0. All the estimates below are made on $[t_0, \varrho)$, which is what makes the inductive hypothesis available rather than assumed: for $s < \varrho$ we have $s - \tau(s) < \varrho$, and either $s - \tau(s) \in [t_0 - T, t_0]$, where the history hypothesis applies, or $s - \tau(s) \in [t_0, \varrho)$, where the definition of ϱ applies; in both cases $\theta(s - \tau(s)) \in \Omega_{2M}$ and hence

$$\|\nabla f(\theta(s))\| \leq G_{2M} \quad \text{and} \quad \|\nabla f(\theta(s - \tau(s)))\| \leq G_{2M} \tag{36}$$

for all $s \in [t_0, \varrho)$. It therefore suffices to show $\mathbb{P}(\varrho \leq t_0 + H \mid \mathcal{F}_{t_0}) \leq \delta$ almost surely.

Partition $[t_0, t_0 + H]$ into $J := \lceil H/h \rceil$ sub-intervals of length $h \in (0, \min\{1, T\}]$ with grid points $s_j := t_0 + jh$. By (2), for any $t \in [s_j, s_{j+1}]$,

$$\theta(t) - \theta(s_j) = -\eta \int_{s_j}^{t} \nabla f(\theta(s - \tau(s)))\, ds + \eta\sigma \left(W(t) - W(s_j)\right). \tag{37}$$

By (36), for $t < \varrho$,

$$\left\|\theta(t) - \theta(s_j)\right\| \leq \eta \left(hG_{2M} + \sigma S_j\right),$$

where

$$S_j := \sup_{u \in [0,h]} \left\|W(s_j + u) - W(s_j)\right\|.$$

Note that Assumptions 2 and 5 guarantee $G_{2M} < \infty$. By the descent lemma for L-smooth functions,

$$\left|f(\theta(t)) - f(\theta(s_j))\right| \leq \left\|\nabla f(\theta(s_j))\right\| \left\|\theta(t) - \theta(s_j)\right\| + \frac{L}{2}\left\|\theta(t) - \theta(s_j)\right\|^2 \leq a\eta R_j + \frac{L}{2}\eta^2 R_j^2, \tag{38}$$

where

$$a := G_{2M}, \quad R_j := hG_{2M} + \sigma S_j.$$

Now, define

$$m(\eta, S) := a\eta(hG_{2M} + \sigma S) + \frac{L}{2}\eta^2 \left(hG_{2M} + \sigma S\right)^2.$$

We have from (38) that, for $s_{j+1} < \varrho$,

$$\sup_{t\in[s_j,s_{j+1}]} V(t) \le V(s_j) + m(\eta, S_j). \tag{39}$$

By Gaussian concentration for the supremum of a k-dimensional Brownian motion, there exist $c_1, c_2 > 0$, depending only on k, such that

$$\mathbb{P}\left(S_j \ge \alpha\sqrt{h}\right) \le c_1 e^{-c_2\alpha^2}, \quad \forall\, \alpha > 0.$$

Choose

$$\alpha := \sqrt{\frac{1}{c_2}\log\left(\frac{c_1 J}{\delta}\right)},$$

so that a union bound over $j = 0, \dots, J-1$ gives $S_j \le \alpha\sqrt{h}$ for all j with probability at least $1-\delta$. Denote this event by $\mathcal{S}$. On $\mathcal{S}$ we have

$$m(\eta, S_j) \le m_\star(\eta) := a\eta\left(hG_{2M} + \sigma\alpha\sqrt{h}\right) + \frac{L}{2}\eta^2\left(hG_{2M} + \sigma\alpha\sqrt{h}\right)^2. \tag{40}$$

We now induct over the sub-intervals to show that, on $\mathcal{S}$, the whole trajectory from t_0 to $t_0 + H$ stays inside Ω_{2M}, i.e., $\varrho > t_0 + H$. Choose

$$\eta \le \bar{\eta} := \min\left\{\frac{M}{4Ja\left(hG_{2M} + \sigma\alpha\sqrt{h}\right)}, \sqrt{\frac{M}{2JL\left(hG_{2M} + \sigma\alpha\sqrt{h}\right)^2}}\right\},$$

so that

$$Jm_\star(\eta) \le \frac{M}{2}.$$

Let

$$V_j := \sup_{t\in[t_0, s_j\wedge\varrho]} V(t), \quad j \in \{0, \dots, J\}.$$

On $\mathcal{S}$, (39) and (40) give

$$\sup_{t\in[s_j, s_{j+1}\wedge\varrho]} V(t) \le V(s_j \wedge \varrho) + m_\star(\eta) \le V_j + m_\star(\eta), \tag{41}$$

and therefore

$$V_{j+1} = \max\left\{V_j, \sup_{t\in[s_j, s_{j+1}\wedge\varrho]} V(t)\right\} \le V_j + m_\star(\eta).$$

Inducting from $V_0 = V(t_0) \le M$ yields $V_j \le M + jm_\star(\eta)$ for all j, and taking $j = J$,

$$\sup_{t\in[t_0,\,(t_0+H)\wedge\varrho]} V(t) \le V_J \le M + Jm_\star(\eta) \le \frac{3M}{2} < 2M$$

on $\mathcal{S}$. If we had $\varrho \le t_0 + H$ then $V(\varrho) = 2M$ by continuity, contradicting the strict inequality above; hence $\varrho > t_0 + H$ on $\mathcal{S}$. Finally, $\mathcal{S}$ is measurable with respect to the Brownian increments after t_0, hence independent of $\mathcal{F}_{t_0}$ (by the strong Markov property when t_0 is a stopping time), so

$$\mathbb{P}\left(\varrho \le t_0 + H \mid \mathcal{F}_{t_0}\right) \le \mathbb{P}\left(\mathcal{S}^c \mid \mathcal{F}_{t_0}\right) = \mathbb{P}(\mathcal{S}^c) \le \delta,$$

which is the desired bound. □

Appendix C: Proof of Theorem 1 Now, in proving the theorem, we split the SDE solution into two phases: the *burn-in* phase and the *local* phase. Intuitively, after initializing θ, say at $\theta(0)$, the burn-in phase corresponds to the regime where the SDE drives θ *rapidly* towards the optimum; in this phase the gradients are relatively large. Once the process enters the local phase, the parameter vector stabilizes near the optimum, so the gradients become correspondingly small. This decomposition allows us to bound the deviation of θ from the optimum in terms of the *local* gradient norm, i.e., the gradient bound valid in the local phase. Without such a refinement, one would need to impose a uniform bound on the gradient norm across all phases, which would in turn yield a larger steady-state noise margin.

We now state the proof formally.

Proof. By Lemma 1 we have $\mathbb{E}[\tau_M] \le T_{\text{burn}} < \infty$, hence $\tau_M < \infty$ almost surely, and $V(u) \le M$ for all $u \in [\tau_M - 2T, \tau_M]$ by (4) and continuity of V. The sublevel set Ω_{2M} is not forward invariant for (2), so we localize: define the exit time

$$\rho := \inf\{s \ge \tau_M : V(s) > 2M\}, \tag{42}$$

and the stopped process $\tilde{\theta}(s) := \theta(s \wedge \rho)$. Every time u appearing below lies in $[\tau_M - 2T, \rho)$: either $u \ge \tau_M$, where $V(u) \le 2M$ by the definition of ρ, or $u \in [\tau_M - 2T, \tau_M)$, where $V(u) \le M$ by (4). The two-delay window of the burn-in time (4) is exactly what makes the second case available. In both cases $\theta(u) \in \Omega_{2M}$, so that

$$\|\nabla f(\theta(u))\| \le G_{2M}, \qquad u \in [\tau_M - 2T,\, \rho), \tag{43}$$

with G_{2M} finite by Assumptions 2 and 5. Applying Lemma 6 at the stopping time τ_M with horizon H and failure level δ – its delayed-history hypothesis being supplied by the definition (4) of τ_M, and its step-size requirement $\eta \le \bar{\eta}$ being part of the hypotheses of the theorem – gives

$$\mathbb{P}\left(\rho \le \tau_M + H \mid \mathcal{F}_{\tau_M}\right) \le \delta \qquad \text{almost surely.} \tag{44}$$

All bounds below are therefore derived for $\tilde{\theta}$ on the horizon $[\tau_M, \tau_M + H]$.

Now we derive an ODE-type inequality on V inside Ω_{2M}. Using Itô's Lemma [12] we have that,

$$dV(t) = -\eta \nabla f(\theta(t))^\top \nabla f(\theta(t-\tau(t)))dt + \eta\sigma \nabla f(\theta(t))^\top dW(t) + \frac{1}{2}\eta^2\sigma^2 \operatorname{tr}[\nabla^2 f(\theta(t))]dt \tag{45}$$

Taking expectations and using $\operatorname{tr}[\nabla^2 f(\theta(t))] \le kL$, where $\theta(t) \in \mathbb{R}^k$, we get

$$\frac{d}{dt}\mathbb{E}[V(t)] \le -\eta \mathbb{E}\left[\nabla f(\theta(t))^\top \nabla f(\theta(t-\tau(t)))\right] + \frac{1}{2}\eta^2\sigma^2 kL, \tag{46}$$

where we have also used the fact that the Brownian term is a martingale and has zero mean. Now we lower bound the inner product $\nabla f(\theta(t))^\top \nabla f(\theta(t-\tau(t)))$. Using the polarization identity

$$a^\top b = \frac{1}{2}\left(\|a\|^2 + \|b\|^2\right) - \frac{1}{2}\|a-b\|^2, \qquad a, b \in \mathbb{R}^k, \tag{47}$$

and taking expectations, we have

$$\begin{aligned}\mathbb{E}[\nabla f(\theta(t))^\top \nabla f(\theta(t-\tau(t)))] =& \frac{1}{2}\mathbb{E}\left[\|\nabla f(\theta(t))\|^2\right] + \frac{1}{2}\mathbb{E}\left[\|\nabla f(\theta(t-\tau(t)))\|^2\right] \\ &- \frac{1}{2}\mathbb{E}\left[\|\nabla f(\theta(t)) - \nabla f(\theta(t-\tau(t)))\|^2\right].\end{aligned}$$

Using (5) to lower bound the first two terms and Lipschitz continuity to upper bound the third term, we get

$$\begin{aligned}\mathbb{E}[\nabla f(\theta(t))^\top \nabla f(\theta(t-\tau(t)))] \ge& \mu\mathbb{E}\left[V(t)\right] + \mu\mathbb{E}\left[V(t-\tau(t))\right] \\ &- \frac{1}{2}L^2\mathbb{E}\left[\|\theta(t) - \theta(t-\tau(t))\|^2\right].\end{aligned} \tag{48}$$

Substituting the bound in (48) into (46) and dropping $-\eta\mu\mathbb{E}[V(t-\tau(t))]$ we conclude that,

$$\frac{d}{dt}\mathbb{E}[V(t)] \le -\eta\mu\mathbb{E}[V(t)] + \frac{\eta L^2}{2}\mathbb{E}\left[\|\theta(t)-\theta(t-\tau(t))\|^2\right] + \frac{\eta^2\sigma^2 kL}{2}. \tag{49}$$

We further bound $\|\theta(t)-\theta(t-\tau(t))\|^2$ as follows:

$$\theta(t)-\theta(t-\tau(t)) = -\eta\int_{t-\tau(t)}^{t}\nabla f(\theta(u-\tau(u)))\,du + \eta\sigma\int_{t-\tau(t)}^{t} dW(u).$$

Using $\|a+b\|^2 \le 2\|a\|^2 + 2\|b\|^2$,

$$\|\theta(t)-\theta(t-\tau(t))\|^2 \le 2\eta^2\left\|\int_{t-\tau(t)}^{t}\nabla f(\theta(u-\tau(u)))\,du\right\|^2 + 2\eta^2\sigma^2\left\|\int_{t-\tau(t)}^{t} dW(u)\right\|^2, \tag{50}$$

we bound each of the two terms above. First we bound the drift term.

$$\begin{aligned}\eta\left\|\int_{t-\tau(t)}^{t}\nabla f(\theta(u-\tau(u)))\,du\right\| &\le \eta\int_{t-\tau(t)}^{t}\|\nabla f(\theta(u-\tau(u)))\|du\\ &\le \eta G_{2M}T.\end{aligned} \tag{51}$$

The noise term admits no pathwise bound, since $\int_{t-\tau(t)}^{t} dW(u) = W(t)-W(t-\tau(t))$ is an unbounded Gaussian vector in $\mathbb{R}^k$; we therefore bound it in conditional expectation. Exactly as in case *(ii)* of the proof of Lemma 3, the delay window of a time $t \in [\tau_M, \tau_M+T)$ straddles the conditioning time τ_M, and the straddling piece $W(\tau_M)-W(t-\tau(t))$ is $\mathcal{F}_{\tau_M}$-measurable rather than centred. With Ψ as in (17), this gives

$$\mathbb{E}\left[\left\|\int_{t-\tau(t)}^{t} dW(u)\right\|^2\middle|\mathcal{F}_{\tau_M}\right] \le kT + \Psi(\tau_M)\mathbb{1}_{\{t<\tau_M+T\}}, \tag{52}$$

the factor k being the dimension of the Brownian motion. Combining the drift and noise parts (51) and (52) and substituting in (50), we obtain, for $t < \rho$,

$$\mathbb{E}\left[\|\theta(t) - \theta(t-\tau(t))\|^2 \,\middle|\, \mathcal{F}_{\tau_M}\right] \le \underbrace{2\eta^2 T^2 G_{2M}^2 + 2k\eta^2\sigma^2 T}_{=:\,\Delta_{\text{local}}} + 2\eta^2\sigma^2\Psi(\tau_M)\mathbb{1}_{\{t<\tau_M+T\}}. \tag{53}$$

As in Remark 10, the boundary term acts only on a layer of length T, is $\mathcal{F}_{\tau_M}$-measurable, and is never exponentiated. Denote

$$\beta_{\text{local}} := \frac{\eta L^2}{2}\Delta_{\text{local}} + \frac{\eta^2\sigma^2 kL}{2}, \quad \text{and} \quad \xi := \eta\mu. \tag{54}$$

It remains to integrate (49) from the *random* time τ_M. Two points require care: the lower limit is a stopping time, and the local bounds hold only up to ρ. We therefore work with

$$\Upsilon(s) := \mathbb{E}\left[V(s\wedge\rho)\,\middle|\,\mathcal{F}_{\tau_M}\right], \qquad s \ge \tau_M.$$

Since $dV(s\wedge\rho) = \mathbb{1}_{\{s<\rho\}}\,dV(s)$ and the segment process $\{\theta(s+u)\}_{u\in[-2T,0]}$ is strong Markov, we may restart (49) at τ_M conditionally on $\mathcal{F}_{\tau_M}$; using (43) and (53) on $\{s<\rho\}$ this gives

$$\frac{d}{ds}\Upsilon(s) \le -\xi\,\mathbb{E}\left[\mathbb{1}_{\{s<\rho\}}V(s)\,\middle|\,\mathcal{F}_{\tau_M}\right] + \beta_{\text{local}} + \eta^3 L^2\sigma^2\Psi(\tau_M)\mathbb{1}_{\{s<\tau_M+T\}}.$$

By continuity $V(\rho) = 2M$, so $\mathbb{E}[\mathbb{1}_{\{s<\rho\}}V(s)\,|\,\mathcal{F}_{\tau_M}] = \Upsilon(s) - 2M\,\mathbb{P}(\rho\le s\,|\,\mathcal{F}_{\tau_M})$, and (44) bounds the latter probability by δ on $[\tau_M, \tau_M+H]$. Hence

$$\frac{d}{ds}\Upsilon(s) \le -\xi\,\Upsilon(s) + \beta_{\text{local}} + 2M\xi\delta + \eta^3 L^2\sigma^2\Psi(\tau_M)\mathbb{1}_{\{s<\tau_M+T\}}.$$

Integrating from $s=\tau_M$ to $s=t$, using $\Upsilon(\tau_M) = V(\tau_M)$ and

$$\int_{\tau_M}^{t} e^{-\xi(t-u)}\,\mathbb{1}_{\{u<\tau_M+T\}}\,du \le T,$$

gives

$$\Upsilon(t) \le (V(\tau_M) - \kappa - 2M\delta)\,e^{-\xi(t-\tau_M)} + \kappa + 2M\delta + \eta^3 L^2\sigma^2 T\,\Psi(\tau_M),$$

where

$$\kappa := \frac{\beta_{\text{local}}}{\xi} = \frac{\eta^2}{\mu}\left(L^2T^2G_{2M}^2 + kL^2\sigma^2T\right) + \frac{\eta\sigma^2 kL}{2\mu}. \tag{55}$$

Noting that $V(\tau_M) \le M$ by (4) and continuity yields the result. □

Appendix D: Proof of Theorem 2

Proof. By Corollary 1 we have $\mathbb{E}[\tau_M] \le T_{\text{burn}} < \infty$, so $\tau_M < \infty$ almost surely and, by Markov's inequality, $\mathbb{P}(\tau_M > t_f) \le T_{\text{burn}}/t_f$; from now on we work on the event $\{\tau_M \le t_f\}$. Let ρ be the exit time (42) and $\tilde{\theta}(s) := \theta(s \wedge \rho)$ as in the proof of Theorem 1. We apply Lemma 6 *once*, at the stopping time τ_M, with horizon t_f and failure level δ; its hypotheses hold as in Appendix C, with $\eta(\cdot) \le \eta_+ \le \bar{\eta}$. This gives $\mathbb{P}(\rho \le t_f \mid \mathcal{F}_{\tau_M}) \le \delta$, and on $[\tau_M - 2T, \rho)$ the bound (43) holds, so that all the local estimates of Appendix C are available on $[\tau_M, t_f]$. A single application is used because the conclusion of Lemma 6 controls V by $2M$ but does not re-establish the hypothesis $V \le M$ needed to restart it; conditioning throughout on $\mathcal{F}_{\tau_M}$, rather than on a later σ-algebra, is what makes the resulting bound $\mathbb{P}(\rho \le t_f \mid \cdot) \le \delta$ usable. We partition $[0, t_f]$ into the blocks $[t_j, t_{j+1}]$, $j \in \{-1, 0, \dots, K-1\}$, of Assumption 6 with $t_{-1} = 0$; these serve only to localize the forcing term β. Itô's Lemma gives, as in (45),

$$dV(t) = -\eta(t)\nabla f(\theta(t))^\top \nabla f(\theta(t - \tau(t)))dt + \eta(t)\sigma \nabla f(\theta(t))^\top dW(t) + \frac{1}{2}\eta(t)^2\sigma^2 \operatorname{tr}[\nabla^2 f(\theta(t))]dt$$

for $t \ge \tau_M$. Taking expectations, the martingale term drops out and we obtain

$$\frac{d}{dt}\mathbb{E}[V(t)] = -\eta(t)\mathbb{E}[\nabla f(\theta(t))^\top \nabla f(\theta(t - \tau(t)))] + \frac{1}{2}\eta(t)^2\sigma^2 \mathbb{E}\left[\operatorname{tr}\left[\nabla^2 f(\theta(t))\right]\right].$$

From here onward we follow exactly along the lines of the proof of Theorem 1: we decompose the inner product using the polarization identity (47), use (5) to lower bound the first two terms and L-smoothness to upper bound the norm difference squared, and drop the non-positive term $-\mu\eta(t)\mathbb{E}[V(t - \tau(t))]$. This yields

$$\frac{d}{dt}\mathbb{E}[V(t)] \le -\mu\eta(t)\mathbb{E}[V(t)] + \frac{1}{2}L^2\eta(t)\mathbb{E}\left[\|\theta(t) - \theta(t - \tau(t))\|^2\right] + \frac{\sigma^2}{2}H_{\max}\eta(t)^2, \tag{56}$$

where $H_{\max} := \sup_{\theta \in \Omega_{2M}} \left|\operatorname{tr}\left(\nabla^2 f(\theta)\right)\right| \le kL$ is deterministic and finite by Assumptions 2 and 5.

It remains to bound the displacement, which for a time-varying learning rate must be done with η evaluated over the whole delay window and not at the endpoint. For $t \in [t_j, t_{j+1}]$ we have, by (2),

$$\theta(t) - \theta(t-\tau(t)) = -\int_{t-\tau(t)}^{t} \eta(u) \nabla f(\theta(u-\tau(u)))\, du + \sigma \int_{t-\tau(t)}^{t} \eta(u)\, dW(u),$$

and $t - \tau(t) \geq t_j - T$, so that $\eta(u) \leq \hat{\eta}_j$ on the window. Proceeding exactly as in (50)–(53), and splitting the stochastic integral at τ_M when the delay window straddles it, the conditional second moment of the Brownian part is at most $k\sigma^2\hat{\eta}_j^2 T + \sigma^2 \Psi_\eta(\tau_M) \mathbb{1}_{\{t<\tau_M+T\}}$, with Ψ_η as in (11). Hence

$$\mathbb{E}\left[\|\theta(t) - \theta(t-\tau(t))\|^2 \,\middle|\, \mathcal{F}_{\tau_M}\right] \leq \underbrace{2\hat{\eta}_j^2 T^2 G_{2M}^2 + 2k\hat{\eta}_j^2 \sigma^2 T}_{=:\, \Delta_{\text{local}}^{(j)}} + 2\sigma^2 \Psi_\eta(\tau_M) \mathbb{1}_{\{t<\tau_M+T\}}. \tag{57}$$

Note that Ψ_η carries the learning rate inside the stochastic integral: for a time-varying η the straddling piece is $\sigma \int_a^{\tau_M} \eta(u) dW(u)$, which is not a multiple of $W(\tau_M) - W(a)$. Substituting (57) into (56) and using $\eta(t) \leq \eta_{\max,\, j} \leq \hat{\eta}_j$ on the block gives

$$\frac{d}{dt}\mathbb{E}[V(t)] \leq -\mu\eta(t)\mathbb{E}[V(t)] + \underbrace{\frac{1}{2}L^2 \Delta_{\text{local}}^{(j)} \eta(t) + \frac{\sigma^2}{2} H_{\max} \eta(t)^2}_{=:\, \beta(t)}, \tag{58}$$

and $\beta(t) \leq \beta_{\max,\, j}$ with $\beta_{\max,\, j}$ as in (10).

This is a linear differential inequality. With the integrating factor $e^{\mu \int_{\tau_M}^t \eta(s) ds}$, and repeating the stopped-process argument of Appendix C verbatim (which contributes the additive terms $2M\delta$ and $\eta_+ L^2 \sigma^2 T\, \Psi_\eta(\tau_M)$), we obtain for $t \in [\tau_M, t_f]$

$$\begin{aligned}\mathbb{E}\left[V(t \wedge \rho) \,\middle|\, \mathcal{F}_{\tau_M}\right] \leq V(\tau_M) e^{-\mu \int_{\tau_M}^t \eta(s) ds} + \int_{\tau_M}^{t} e^{-\mu \int_u^t \eta(s) ds} \beta(u)\, du \\ + 2M\delta + \eta_+ L^2 \sigma^2 T\, \Psi_\eta(\tau_M).\end{aligned} \tag{59}$$

It remains to bound the residual integral in (59) by a constant. By the definition of η_- in Assumption 6, $\eta(s) \geq \eta_-$ for all $s \in [0, t_f]$, hence

$$e^{-\mu \int_u^t \eta(s) ds} \leq e^{-\mu\eta_-(t-u)}, \qquad 0 \leq u \leq t \leq t_f. \tag{60}$$

Splitting the integral over the blocks and bounding β by $\beta_{\max,j}$ on each of them,

$$\begin{aligned}\int_{\tau_M}^{t} e^{-\mu\int_u^t \eta(s)ds}\beta(u)\,du &= \sum_{j=-1}^{K-1}\int_{t_j\vee\tau_M}^{t_{j+1}\wedge t} e^{-\mu\int_u^t \eta(s)ds}\beta(u)\,du \\ &\le \sum_{j=-1}^{K-1}\beta_{\max,j}\int_{t_j\vee\tau_M}^{t_{j+1}\wedge t} e^{-\mu\eta_-(t-u)}du, \end{aligned} \tag{61}$$

with the convention that an interval with $t_j\vee\tau_M > t_{j+1}\wedge t$ contributes nothing. For each j, writing $A_j := t_j\vee\tau_M$ and $B_j := t_{j+1}\wedge t$,

$$\int_{A_j}^{B_j} e^{-\mu\eta_-(t-u)}du = \frac{e^{-\mu\eta_-(t-B_j)} - e^{-\mu\eta_-(t-A_j)}}{\mu\eta_-} \le \frac{1-e^{-\mu\eta_-\left(t_{j+1}-t_j\right)}}{\mu\eta_-}, \tag{62}$$

where the last step uses $e^{-\mu\eta_-(t-B_j)}\le 1$, factors out the common exponential, and uses $B_j - A_j \le t_{j+1}-t_j$. Summing over j and substituting into (61) bounds the residual by $\kappa(K)$ as defined in (9), which together with (59) yields the result. The bound is uniform in t, so $\kappa(K)$ is a genuine constant margin. If instead one retains the first part of (62), the sum telescopes whenever the $\beta_{\max,j}$ are equal – in particular for a constant learning rate – and gives $\kappa\left(1-e^{-\eta\mu(t-\tau_M)}\right)$, recovering Theorem 1.

□

We close with two remarks on the step size. The first records, as an aside, for which step sizes the right-hand side of (56) is strictly negative; this is an interpretation of the bound rather than a hypothesis of Theorem 2, since the threshold below depends on the very quantity $\mathbb{E}[V(t)]$ that is being estimated.

REMARK 13. (**Step sizes giving strictly negative drift**): Substituting (57) into (56) with $\hat{\eta}_j$ replaced by the running value $\eta(t)$, the right-hand side becomes the cubic

$$q(\eta(t)) := \eta(t)\left[a\,\eta(t)^2 + \frac{\sigma^2 H_{\max}}{2}\eta(t) - \mu\mathbb{E}[V(t)]\right], \tag{63}$$

where

$$a := L^2T^2G_{2M}^2 + kL^2\sigma^2T.$$

Its roots are $\eta_1(t)=0$ and

$$\eta_2(t) = \frac{-\sigma^2H_{\max} + \sqrt{\sigma^4H_{\max}^2 + 16\mu\mathbb{E}[V(t)]a}}{4a}, \tag{64a}$$

$$\eta_3(t) = \frac{-\sigma^2 H_{\max} - \sqrt{\sigma^4 H_{\max}^2 + 16\mu\mathbb{E}[V(t)]a}}{4a} < 0. \tag{64b}$$

Whenever $\mathbb{E}[V(t)] > 0$, the bracket in (63) is an upward parabola which is negative at $\eta(t) = 0$; hence, for $\eta(t) > 0$, $q(\eta(t)) < 0$ precisely when $0 < \eta(t) < \eta_2(t)$. •

REMARK 14. (**Delay-free limit**): In the absence of delay ($T = 0$) we have $a = 0$, so (63) degenerates to a quadratic and the condition for strictly negative drift becomes

$$0 < \eta(t) < \frac{2\mu\mathbb{E}[V(t)]}{\sigma^2 H_{\max}}.$$

This is consistent with (64a): expanding the square root for small a gives $\eta_2(t) \to 2\mu\mathbb{E}[V(t)]/(\sigma^2 H_{\max})$ as $a \to 0$. The same threshold is obtained directly by solving

$$-\mu\eta(t)\mathbb{E}[V(t)] + \frac{\sigma^2}{2} H_{\max}\eta(t)^2 < 0.$$

•

Appendix E: Supplementary Proofs Recall from Section 6 that the synthetic function we chose for our experiments was:

$$f(\theta) = \theta^\top Q\theta + \epsilon \sin(w^\top\theta). \tag{65}$$

Throughout, $Q = Q^\top \succ \mathbf{0}$, $\epsilon \geq 0$ and $w \neq \mathbf{0}$. In what follows, we derive admissible values for the strong-convexity modulus μ and the L-smoothness constant L for this function; both are bounds on the extreme eigenvalues of the Hessian, uniformly in θ, which is what the analysis requires.

E.1. Derivation of strong-convexity constant Define

$$R_Q(v) := \frac{v^\top Q v}{\|v\|^2},$$

to be the Rayleigh quotient of the symmetric positive-definite matrix Q at a vector $v \neq \mathbf{0}$. The gradient and Hessian of f are given respectively as:

$$\nabla f(\theta) = 2Q\theta + \epsilon\cos(w^\top\theta)w, \tag{66a}$$

$$\nabla^2 f(\theta) = 2Q - \epsilon\sin(w^\top\theta)ww^\top. \tag{66b}$$

Pick any unit vector v. Using,

$$v^\top(ww^\top)v = (v^\top w)^2,$$

we have from (66b),

$$v^\top \left(\nabla^2 f(\theta)\right) v = 2R_Q(v) - \epsilon \sin(w^\top \theta)(v^\top w)^2.$$

Now using, $R_Q(v) \geq \lambda_{\min}(Q)$, and $\sin(\cdot) \in [-1, +1]$, we can lower-bound $v^\top(\nabla^2 f(\theta))v$ as,

$$v^\top \left(\nabla^2 f(\theta)\right) v \geq 2\lambda_{\min}(Q) - \epsilon(v^\top w)^2 \geq 2\lambda_{\min}(Q) - \epsilon\|w\|^2,$$

where the last inequality follows from the Cauchy–Schwarz inequality and $\|v\| = 1$. Therefore, provided that

$$\epsilon < \frac{2\lambda_{\min}(Q)}{\|w\|^2},$$

the Hessian is uniformly positive definite and f is μ-strongly convex with

$$\mu := 2\lambda_{\min}(Q) - \epsilon\|w\|^2.$$

E.2. Derivation of L-smoothness constant Now we upper bound $v^\top(\nabla^2 f(\theta))v$, which, in a similar fashion to the lower bound, satisfies

$$v^\top(\nabla^2 f(\theta))v \leq 2\lambda_{\max}(Q) + \epsilon\|w\|^2.$$

Combining this with the lower bound of the previous subsection, and using the fact that $-\lambda_{\max}(Q) \leq \lambda_{\min}(Q)$, gives the two-sided estimate

$$\left|v^\top(\nabla^2 f(\theta))v\right| \leq 2\lambda_{\max}(Q) + \epsilon\|w\|^2,$$

for every unit vector v. Since $\nabla^2 f(\theta)$ is symmetric, its operator norm therefore satisfies

$$\left\|\nabla^2 f(\theta)\right\|_2 \leq L := 2\lambda_{\max}(Q) + \epsilon\|w\|^2,$$

for all θ, so f is L-smooth.